

# Storyboard: Bridging the Gap: From Innovation to Patient Impact

Ethics, Evidence, Regulation and Responsible Medical Device Translation • Dr Vikas Kumar Jha • KIIT University, Bhubaneswar • 40-min core 39:30 • full 70:00

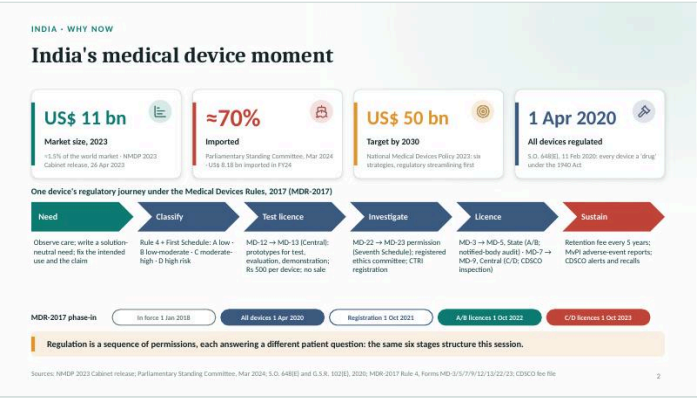
**Story spine.** Two composite patients, Ramesh (Parkinson's, freezing of gait, FoGO) and Kamala (knee osteoarthritis, SwaKnee), followed along six stages: Need → Evidence → People → Permission → Access → Safety. Permission is the deep module: regulation in India by risk class, CDSCO obligations and guidance, then the comparison abroad.

**Two paths.** 40-minute workshop: the 24 slides flagged core (39.50 min scripted): every vote and decision, the FoGO and SwaKnee evidence, risk classes, the licence matrix, FoGO's path, the quiz, the world map and the India-USA-EU comparison • 75-minute seminar: all 54 main slides (70 min scripted), with the full regulatory deep-dive

Evidence key **Demonstrated** **Reported, preliminary** **Planned** **Not established**



SLIDE 1 • OPENING • 0:00–0:30 • 0.5 MIN • **CORE (40-MIN PATH)**  
**Title with authentic device photographs**  
**Visual** Gradient title layout; FoGO and SwaKnee rounded photos with labels; presenter block  
**Interaction** —



SLIDE 2 • OPENING • 0:30–2:00 • 1.5 MIN • **CORE (40-MIN PATH)**  
**A fast-growing, import-dependent market that only recently brought every device under regulation**  
**Visual** statTiles (market size, import share, 2030 ambition, all devices regulated 2020) + chevronFlow of the regulatory journey + callout  
**Interaction** —

DISCLOSURE

### My interests, stated first

Interests in the two case devices

**FoGO • Ahilaya Biomedicals Pvt Ltd**  
Fretting-of-gait wearable - incubated at KIT TüB

Alpha prototype - HD-13

- Founder and principal investigator
- Provisional patent 20231119165 and a design application filed
- ₹25.5 lakh in grants: Startup Odisha • DST NIDH-FRAXAS • Startup India Seed Fund
- BIRAC Biotechnology Ignition Grant proposal under review (BIG: up to ₹50 lakh over 18 months)

**SwaKnee • Swayogya Rehab Solutions**  
Pulsed electromagnetic field (PEMF) knee system

[state role and any equity, royalty or salary interest]

Three ground rules for today

- Vote before you hear my view**  
Record your intuition first; we compare it with the closing vote on the same options.
- Challenge every claim, including mine**  
Ask of each claim: what is the evidence, who measured it, and who paid for it?
- Protect identities**  
No names or photographs of patients from your own practice; personas here are composites.

A conflict of interest is a set of circumstances that creates a risk that judgement about a primary interest will be unduly influenced by a secondary interest (Institute of Medicine, 2009); a risk, not an accusation. Test everything I say by whether a conflict would be reasonably perceived (University of Cambridge policy).

Sources: Ahilaya Biomedicals project material (supplied); BIRAC BIG scheme; Institute of Medicine 2009; University of Cambridge COI policy

SLIDE 3 • OPENING • 2:00–3:00 • 1.0 MIN • **CORE (40-MIN PATH)**

## Model the COI practice we will teach

Visual Interest panel (FoGO: founder/PI, patent, grants, BIRAC proposal; SwaKnee: [state role]) + three ground rules as icon rows

Interaction —

OPENING VOTE

### Would you let a patient use it?

Three facts about one device (not named yet)

- A working prototype
- Promising results on public datasets
- A test licence granted

Record the split on the flipchart

1 =  2 =  3 =

Hold up one, two or three fingers, then tell your neighbour why. We vote again at the end with exactly the same options.

Three options (one, two or three fingers)

- Routine care**  
Suitable for a clinician to recommend to patients now
- Research study**  
Suitable for a study with ethics approval and consent
- Not yet**  
I need more information before either

Sources: MDR 2017 (frames HD-12/13, HD-22/23); NDCI Rules 2019; project material (supplied, 2020)

SLIDE 4 • OPENING • 3:00–5:00 • 2.0 MIN • **CORE (40-MIN PATH)**

## Three facts, three options; record the split

Visual factRows + optionCards (1 Routine care, 2 Research study, 3 Not yet)

Interaction Vote by fingers; flipchart tally

LENSES

### Three teaching traditions, one Indian patient

**Stanford Biodesign** **NEED**

- Identify → Invent → Implement: start from a validated need, not a technology
- Need Statements (Student Guide, 2022): solution-neutral, screened before inventing
- Privileged Design-Making (2022): write the team's ethical principles early
- India: Stanford-India Biodesign since 2007 (32 fellows, 14 technologies, 12 companies); now the School of International Biodesign at AIIMS; an AIIMS emergency-department study filtered 100 needs to 10 (Chaturvedi 2015)

**Harvard research ethics** **PEOPLE**

- HMS Center for Bioethics: case-based teaching of research ethics
- MBCT Center (Brigham and Harvard): Diversity guidance (2020), Accessibility by Design toolkit (2023), plain-language glossary and trial access for devices (2025)
- Petrie-Fram Center: Diagnosing in the Home; ethics of remote device monitoring (Cohen, Gerke & Kramer 2020)
- HMS conflict-of-interest policy (2010): disclose, then manage

**Cambridge design and translation** **ACCESS • SAFETY**

- Engineering Design Centre Inclusive Design Toolkit and Exclusion Calculator: who is excluded by a product's demands?
- Engineering Better Care (2017): working group chaired by Prof John Clarkson: people, systems, design and risk, with a 'Sustain' phase
- Institute for Biomedical Innovation (launched January 2020): From lab prototype to trial-ready device
- Cambridge Judge CODE: making proven solutions affordable

None of these institutions has evaluated or endorsed FoGO or SwaKnee. The Biodesign textbook is Stanford content published by Cambridge University Press.

Applied to Indian rules and Indian patients

**CDSCO • MDR 2017**  
Permeation Rule 4 (Gases A-D); Biocompatibility Rule 5 (MDR-9); and Licence MD-13

**ICMR 2017**  
Propose: National Ethical Guidelines; ethics committees registered under NDCI Rules 2019

**MvPI • IPC**  
Safety: adverse-event reporting since 6 Jul 2015; 174 monitoring centres

**NPPA • DPCO 2013**  
Across: ceiling prices on sterile and knee implants (2017); trade-margin caps (2021)

**BIS**  
Standards: Indian Standards Essential Principles checklist (Class A self-certified)

**DPDP Rules 2025**  
Data: personal health data from apps and cloud; phased commencement

Sources: Stanford Biodesign guides (2022); MBCT Center; Petrie-Fram; Cambridge EDC; Engineering Better Care (2017); MDR 2017; ICMR 2017; IPC MvPI; NPPA (PH 2017, 2021); DPDP Rules 2025

SLIDE 5 • OPENING • 5:00–5:45 • 0.75 MIN • **DEEP-DIVE (75-MIN PATH)**

## Stanford Biodesign, Harvard research ethics, Cambridge design; adapted to Indian rules

Visual cardGrid 3 (named materials per tradition) + India adaptation band (CDSCO, ICMR, MvPI, NPPA, BIS, NPPA) with icons

Interaction —

PATIENTS

Two people we will follow

Ramesh and Kamala are composites, not real patients, built from situations common in Odisha clinics. Every decision today is made for one of them: when you vote, vote for them.

**Ramesh, 71**  
Parkinson's disease with freezing of gait

**≈7.7 lakh**  
people living with Parkinson's disease in India, 2019  
India State Level Disease Burden Initiative (ISDBI 2019), as cited in the FoGO project material

- Parkinson's disease for eight years
- Freezes in doorways and when turning
- Has fallen twice this year
- Lives with his daughter, who works days and lends him her smartphone

Device in this story: **FoGO, freezing-of-gait wearable**    **Alpha prototype - MD-13**

**Kamala, 64**  
Knee osteoarthritis in both knees

**28.7%**  
knee osteoarthritis in a five-site community survey  
(n = 5,000)  
Pat et al., Indian J Orthop 2016 (one study, not national)

- Osteoarthritis in both knees
- Lives 30 km from the clinic at the district hospital
- Travels by shared auto; someone must accompany her
- Household pays out of pocket and loses wages on clinic days

Device in this story: **SwaKnee, PEXAF knee system**    **Product - 45-min daily sessions**

Sources: composite personas (workshops); India State Level Disease Burden Initiative; Lancet Glob Health 2021; Pat et al., Indian J Orthop 2016; SwaKnee ktrafit (supplied)

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SLIDE 6 • OPENING • 5:45–6:45 • 1.0 MIN • **CORE (40-MIN PATH)**

# Ramesh and Kamala: composites with real burdens

**Visual** Two persona cards with photo (FoGO ankle in use; SwaKnee applicator), facts, and a context stat each (PD burden; knee OA prevalence)

**Interaction** —

THE JOURNEY

Every stage answers a patient question

**Need**  
"Will it help me?"

1

Q

**Observe** care, write a solution-neutral need, state the intended use and the claim. The School of International Biodesign (AIMS, IIT Delhi) teaches the method in India.

**Evidence**  
"Is there proof, for people like me?"

2

A

**Clinical investigation** permission MD-22 → MD-23 (seventh Schedule documents); prospective CTN registration; ISO 14155 good clinical practice.

**People**  
"Can I freely say no?"

3

P

**Ethics committee** registered under the NDCT Rules 2019; ICMR 2017 guidelines on consent (s.5), vulnerability (s.8); device trials (s.7.7); interests disclosed.

**Permission**  
"Is it allowed for this use?"

4

D

**CDSCO - State licensing pathway**

**Manufacturing licence** MD-5 (Class A/B, State) or MD-9 (Class C/D, Central); test licence MD-13 for prototypes; import licence MD-15 for any class.

**Access**  
"Can I afford it and keep using it?"

5

A

**NPPA - timeline**

**DPCO 2013** ceiling price; **concomitant items** (Feb 2017); **knee implants** (Aug 2017, cap continued to 15 Nov 2020); most outpatient devices are paid for out of pocket.

**Safety**  
"Who answers if it fails?"

6

S

**MvPI at IPC - CDSCO**

**Adverse event** reporting since 6 Jul 2015 through 174 monitoring centres; CDSCO alerts, recalls and licence cancellation (DePay A&R hp, 2013).

The six dots at the top of later slides show which stage we are in. Stages are not strictly sequential: Engineering Better Care's 'Sustain' phase means access and safety are designed in from the start, not added after launch.

Sources: ICMR 2017; MDR 2017 Forms MD-5/9/13/15/22/23; NDCT Rules 2019; ISO 14155; NPPA; IPH 2017; cap to 15 Nov 2020; IPC; MvPI; CDSCO A&R alert 2013; Engineering Better Care 2017

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SLIDE 7 • OPENING • 6:45–7:30 • 0.75 MIN • **DEEP-DIVE (75-MIN PATH)**

# Six stages, six questions; the tracker dots follow them

**Visual** Six icon discs on a line with question + the Indian permission that answers each (e.g., Permission → CDSCO; Access → NPPA; Safety → MvPI)

**Interaction** —

01

PART 1 • NEED AND EVIDENCE

What are we promising, and what has been shown?

Two devices, two patients. First the wording of the need and the claim; then what each prototype has demonstrated, reported, planned or not yet established.

- A solution-neutral need, and a claim ladder that sets the evidence bar
- Is it a medical device in India? The intended purpose decides, not the technology
- FoGO and SwaKnee on the evidence ladder, with a decision to vote on

FoGO alpha prototype (FoGO 2020) phone app, chest module on an adhesive patch, ankle module

SLIDE 8 • NEED AND EVIDENCE • 7:30–7:45 • 0.25 MIN • **DEEP-DIVE (75-MIN PATH)**

# What are we promising, and what has been shown?

**Visual** sectionDivider with FoGO system photo

**Interaction** —


**NEED · WORDING**

# Wording sets the evidence bar


**Stanford Biodesign need statement**

A way to **address the problem** [in a **specific population**] in order to **[achieve a desired outcome]**

**Ramsey:** a way to reduce freezing-related immobility and falls in people with Parkinson's disease living at home, in order to **prevent safe, independent walking**.

No sensor, no vibrator, no app, the need stays solution-neutral.

**Your turn (60 s):** rewrite Kameda's need without naming any device.

**Why the words matter:** Under MDR 1710 the intended use and the claims describe a **medical device** and which **falls in Class IIa** in CE marking software guidance (21 NE 2020) grants the **risk** by how far the output drives clinical decisions and thus serves the condition.

**INFORMATION**

- Evidence needed
- Benchmark
- Feasibility
- Permission in India
- No medical purpose outside MDR 1710
- Real purpose, doesn't need the label

**MONITORING**

"Thanks you for your neurologist!"

- Evidence needed
- Analytical validation of the device
- Software IVD, cybersecurity
- Stability in total clinics
- Permission in India
- Medical purpose a device
- GDPR rules by DSGVO guidelines 2017
- License MDD 15 (class II) or MDD 17 (medical)

**DETECTION**

"Detects freezing of gait in Parkinson's"

- Evidence needed
- Sensitivity, specificity vs. video labeled freezes
- Falsc alarm per hour, strategy
- Alarm, haptic, displays, usability, etc.
- Remotely IVD
- MD 10 + MD 12
- MD 13
- Clinical investigation
- MD 12 + MD 23
- Registered ethics
- Then, the CE

**TREATMENT**

"Reduces freezing and falls"

- Evidence needed
- Prospective controlled trial falls, walking
- Usability and safety
- Permeation in India
- Permeation in all
- All of the intervention range
- Clinical context
- UKCA2020, 2020, CHINA
- MDA, IVD
- Permit for sale
- signage after launch

**Stronger claim → more evidence, more permissions, closer scrutiny**

Sources: Stanford Biodesign Need Statements (2022); MDR-2017; CISO2 MDSW guidance 21 NE 2020; Formis MDD-12/12; MDD-22/22; UKCA2020; CHINA 1954; Simon et al., JAMA 2022

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SLIDE 9 • NEED AND EVIDENCE • 7:45-9:15 • 1.5 MIN • **CORE (40-MIN PATH)**

## Solution-neutral need; claim wording decides obligations

**Visual** Stanford need-statement card with Ramesh example + a 4-rung claim ladder (information → monitoring → detection → treatment) with the evidence and licence each needs

### Interaction Rewrite Kamala's need

**NEED - IS IT A DEVICE?**

# Is it a Medical Device in India?

### 1 - What is it for?

**Medical purpose**

CDSCO Act 1984, 13(b)(1), devices intended for use on human beings for the diagnosis, treatment, mitigation or prevention of disease or disorder.

All such devices were notified on 11 Feb 2020 (G.O. 44883) and registered from 1 Apr 2020.

### 2 - Software doing medical work?

saMD or software in a device: CDSCO guidance CDSCO/MSDW/REGD/20/2024 (11 Feb 2024). Class follows the medical purpose. Not for the original device design and development, and the security of the condition, data and IOT included.

### 3 - Which risk class?

Rule 4 and First Schedule Part I: Non-invasive, invasive (surface or surgical), implant, short-term, long-term, active (energy, diagnosis, diagnosis and control), and special risks - Class A (low), Class B (high).

### 4 - Regulated device

Class A/S: Low risk devices  
Class B: Non-invasive  
Class C: Invasive  
Class D: Invasive  
Class E: Invasive  
Class F: Invasive  
Class G: Invasive  
Class H: Invasive  
Class I: Invasive  
Class J: Invasive  
Class K: Invasive  
Class L: Invasive  
Class M: Invasive  
Class N: Invasive  
Class O: Invasive  
Class P: Invasive  
Class Q: Invasive  
Class R: Invasive  
Class S: Invasive  
Class T: Invasive  
Class U: Invasive  
Class V: Invasive  
Class W: Invasive  
Class X: Invasive  
Class Y: Invasive  
Class Z: Invasive  
Class AA: Invasive  
Class AB: Invasive  
Class AC: Invasive  
Class AD: Invasive  
Class AE: Invasive  
Class AF: Invasive  
Class AG: Invasive  
Class AH: Invasive  
Class AI: Invasive  
Class AJ: Invasive  
Class AK: Invasive  
Class AL: Invasive  
Class AM: Invasive  
Class AN: Invasive  
Class AO: Invasive  
Class AP: Invasive  
Class AQ: Invasive  
Class AR: Invasive  
Class AS: Invasive  
Class AT: Invasive  
Class AU: Invasive  
Class AV: Invasive  
Class AW: Invasive  
Class AX: Invasive  
Class AY: Invasive  
Class AZ: Invasive  
Class BA: Invasive  
Class BB: Invasive  
Class BC: Invasive  
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Class BE: Invasive  
Class BF: Invasive  
Class BG: Invasive  
Class BH: Invasive  
Class BI: Invasive  
Class BJ: Invasive  
Class BK: Invasive  
Class BL: Invasive  
Class BM: Invasive  
Class BN: Invasive  
Class BO: Invasive  
Class BP: Invasive  
Class BQ: Invasive  
Class BR: Invasive  
Class BS: Invasive  
Class BT: Invasive  
Class BU: Invasive  
Class BV: Invasive  
Class BW: Invasive  
Class BX: Invasive  
Class BY: Invasive  
Class BZ: Invasive  
Class CA: Invasive  
Class CB: Invasive  
Class CC: Invasive  
Class CD: Invasive  
Class CE: Invasive  
Class CF: Invasive  
Class CG: Invasive  
Class CH: Invasive  
Class CI: Invasive  
Class CJ: Invasive  
Class CK: Invasive  
Class CL: Invasive  
Class CM: Invasive  
Class CN: Invasive  
Class CO: Invasive  
Class CP: Invasive  
Class CQ: Invasive  
Class CR: Invasive  
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Class DO: Invasive  
Class DP: Invasive  
Class DQ: Invasive  
Class DR: Invasive  
Class DS: Invasive  
Class DT: Invasive  
Class DU: Invasive  
Class DV: Invasive  
Class DW: Invasive  
Class DX: Invasive  
Class DY: Invasive  
Class DZ: Invasive  
Class EA: Invasive  
Class EB: Invasive  
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Class EF: Invasive  
Class EG: Invasive  
Class EH: Invasive  
Class EI: Invasive  
Class EJ: Invasive  
Class EK: Invasive  
Class EL: Invasive  
Class EM: Invasive  
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Class EO: Invasive  
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Class ET: Invasive  
Class EU: Invasive  
Class EV: Invasive  
Class EW: Invasive  
Class EX: Invasive  
Class EY: Invasive  
Class EZ: Invasive  
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Class FB: Invasive  
Class FC: Invasive  
Class FD: Invasive  
Class FE: Invasive  
Class FF: Invasive  
Class FG: Invasive  
Class FH: Invasive  
Class FI: Invasive  
Class FJ: Invasive  
Class FK: Invasive  
Class FL: Invasive  
Class FM: Invasive  
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Class FO: Invasive  
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Class FR: Invasive  
Class FS: Invasive  
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Class IC: Invasive  
Class ID: Invasive  
Class IE: Invasive  
Class IF: Invasive  
Class IG: Invasive  
Class IH: Invasive  
Class II: Invasive  
Class IJ: Invasive  
Class IK: Invasive  
Class IL: Invasive  
Class IM: Invasive  
Class IN: Invasive  
Class IO: Invasive  
Class IP: Invasive  
Class IQ: Invasive  
Class IR: Invasive  
Class IS: Invasive  
Class IT: Invasive  
Class IU: Invasive  
Class IV: Invasive  
Class IW: Invasive  
Class IX: Invasive  
Class IY: Invasive  
Class IZ: Invasive  
Class JA: Invasive  
Class JB: Invasive  
Class JC: Invasive  
Class JD: Invasive  
Class JE: Invasive  
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Class JH: Invasive  
Class JI: Invasive  
Class JJ: Invasive  
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Class JP: Invasive  
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Class OK: Invasive  
Class OL: Invasive  
Class OM: Invasive  
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Class OV: Invasive  
Class OW: Invasive  
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Class PG: Invasive  
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Class PO: Invasive  
Class PP: Invasive  
Class PQ: Invasive  
Class PR: Invasive  
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Class PV: Invasive  
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Class PX: Invasive  
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Class QM: Invasive  
Class QN: Invasive  
Class QO: Invasive  
Class QP: Invasive  
Class QQ: Invasive  
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Class QT: Invasive  
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Class QV: Invasive  
Class QW: Invasive  
Class QX: Invasive  
Class QY: Invasive  
Class QZ: Invasive  
Class RA: Invasive  
Class RB: Invasive  
Class RC: Invasive  
Class RD: Invasive  
Class RE: Invasive  
Class RF: Invasive  
Class RG: Invasive  
Class RH: Invasive  
Class RI: Invasive  
Class RJ: Invasive  
Class RK: Invasive  
Class RL: Invasive  
Class RM: Invasive  
Class RN: Invasive  
Class RO: Invasive  
Class RP: Invasive  
Class RQ: Invasive  
Class RR: Invasive  
Class RS: Invasive  
Class RT: Invasive  
Class RU: Invasive  
Class RV: Invasive  
Class RW: Invasive  
Class RX: Invasive  
Class RY: Invasive  
Class RZ: Invasive  
Class SA: Invasive  
Class SB: Invasive  
Class SC: Invasive  
Class SD: Invasive  
Class SE: Invasive  
Class SF: Invasive  
Class SG: Invasive  
Class SH: Invasive  
Class SI: Invasive  
Class SJ: Invasive  
Class SK: Invasive  
Class SL: Invasive  
Class SM: Invasive  
Class SN: Invasive  
Class SO: Invasive  
Class SP: Invasive  
Class SQ: Invasive  
Class SR: Invasive  
Class SS: Invasive  
Class ST: Invasive  
Class SU: Invasive  
Class SV: Invasive  
Class SW: Invasive  
Class SX: Invasive  
Class SY: Invasive  
Class SZ: Invasive  
Class TA: Invasive

SLIDE 10 • NEED AND EVIDENCE • 9:15–10:30 • 1.25 MIN • DEEP-DIVE (75-MIN PATH)

## The intended purpose, not the technology, decides

**Visual** Decision tree (D&C Act s.3(b)(iv) definition; software; wellness vs medical) with FoGO, SwaKnee, a fitness band and a hospital app as worked examples

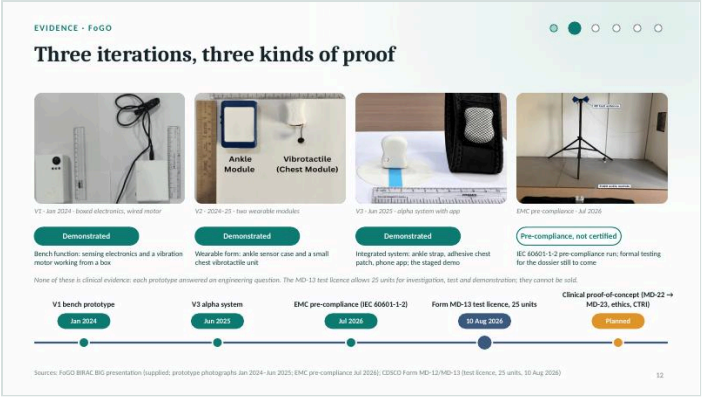
### Interaction —

[illegible]SLIDE 11 • NEED AND EVIDENCE • 10:30–12:00 • 1.5 MIN • **CORE (40-MIN PATH)**

## Seen vs not yet shown

**Visual** Embedded 12 s clip + system diagram (ankle module → app → chest module, 100 Hz, 255 ms) + Seen/Not yet shown columns

**Interaction** What did you see? What did you infer?

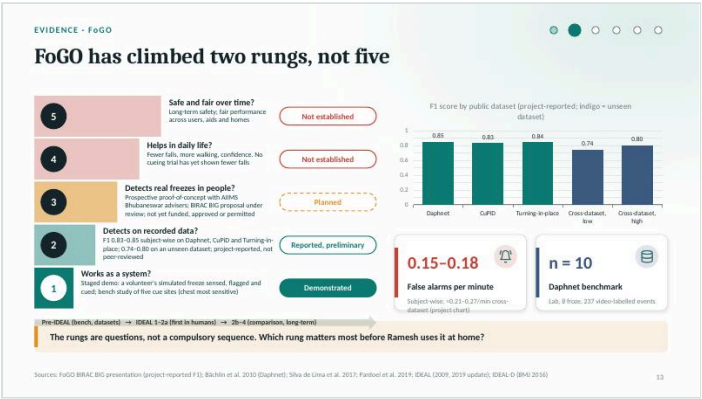


SLIDE 12 • NEED AND EVIDENCE • 12:00–13:00 • 1.0 MIN • DEEP-DIVE (75-MIN PATH)

## Each prototype answered a different question; none is clinical evidence

Visual Photo strip v1→v2→v3 (authentic) with dates and what each proved, status chips, plus EMC test photo and MD-13 milestone

Interaction —

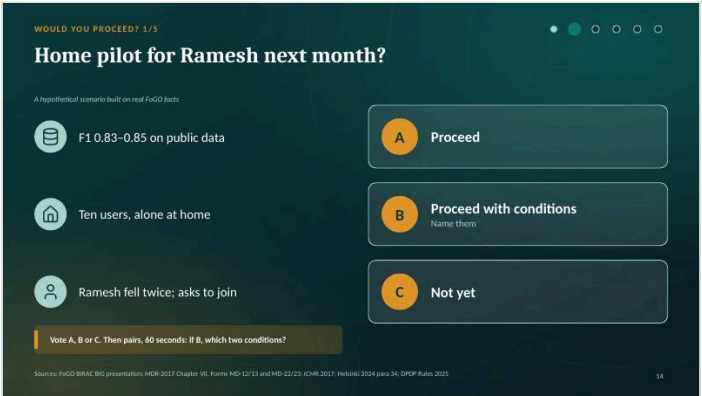


SLIDE 13 • NEED AND EVIDENCE • 13:00–14:30 • 1.5 MIN • CORE (40-MIN PATH)

## Demonstrated / Reported / Planned / Not established on the IDEAL ladder

Visual classLadder-style five-rung evidence staircase + native bar chart of F1 by dataset (Daphnet 0.85, CuPiD 0.83, Turning 0.84; cross-dataset 0.74–0.80)

Interaction —



SLIDE 14 • NEED AND EVIDENCE • 14:30–17:30 • 3.0 MIN • CORE (40-MIN PATH)

## Permissions first, then a supervised study

Visual factRows + optionCards

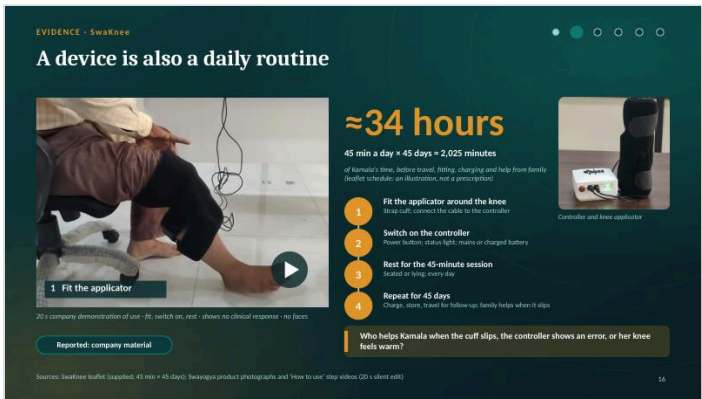
Interaction Vote; pairs name two conditions



SLIDE 15 · NEED AND EVIDENCE · 17:30-18:45 · 1.25 MIN · DEEP-DIVE (75-MIN PATH)

## ISO 14971 thinking: estimate, control, verify, monitor

Visual 3x3 risk matrix with six FoGO failure modes + '9-16 false cues per walking hour' stat  
Interaction Which control first?



SLIDE 16 · NEED AND EVIDENCE · 18:45-20:00 · 1.25 MIN · DEEP-DIVE (75-MIN PATH)

## Use burden is part of benefit-risk

Visual Embedded 20 s clip + controller/applicator photo + '≈34 hours' stat + routine steps  
Interaction —



SLIDE 17 · NEED AND EVIDENCE · 20:00-22:30 · 2.5 MIN · CORE (40-MIN PATH)

## Company-reported average vs patient-important change

Visual Native bar chart (32% vs 14%) + MCII callout + three claim cards with status chips + 'what the summary does not report' list

Interaction Defend claim 1, challenge claim 3

EVIDENCE • INDEPENDENT

What independent evidence says about cueing and PEMF

FoGO domain - detecting and cueing freezing of gait

Enlighten benchmark (Blaich 2018): 10 patients in a laboratory, 8 from 227 video-labelled freezes

Benchmark, n = 10

Silva de Lima 2017, systematic review: sensitivity 73–100%, specificity 67–100%, mostly laboratory studies

100% observed

RESCUE trial (n = 153, home cueing): small gait gains and lower freezing severity in knees; fall counts not measured

Falls not measured

Glinh 2018 narrative review: no trial found showing lower falls from cueing, continuous or on-demand

Fewer falls: not shown

FoGO: project-reported F1 on public datasets; prospective study planned, not done

Expected, preliminary

SwaKnee domain - PEMF for knee osteoarthritis

Cochrane 2013 (9 trials, 636 adults, all OA sites and field types): pain probably improves ~15/100 more than sham; function uncertain

Moderate: pain only

Chen 2019 (knee, 8 RCTs): no pain advantage vs placebo; Yang 2020 (all sites, 16 RCTs): an advantage

Review ongoing

OMSI 2019 guideline: electromagnetic therapy strongly recommended against (low-quality evidence)

Strongly against

NICE NG224 (2022): do not offer listed electrotherapies; insufficient evidence of benefit (list does not name PEMF)

Insufficient evidence

Cartilage regrowth: no human evidence found in any review; pooled outcomes are pain, stiffness, function

Not established

Sources: Blaich 2018; Silva de Lima 2017; Newberry 2007 (RESCUE); Glinh 2018; Li et al., Cochrane 2013; Chen 2019; Yang 2020; Markovic 2022; OMSI 2019; NICE NG224 (2022)

18

SLIDE 18 • NEED AND EVIDENCE • 22:30–23:30 • 1.0 MIN • DEEP-DIVE (75-MIN PATH)

## Mixed and modest; no cartilage regrowth evidence

Visual Two-column evidence panel (FoGO domain; SwaKnee domain) with icon rows and a GRADE-style certainty pill per finding

Interaction —

EVIDENCE • EQUITY

An average can hide a patient

Arterial saturation below 88% while the oximeter read 92–94% (% of paired readings)

11.7%

3.6%

Black patientsWhite patients

Spindig et al., NEJM 2020: 86 of 148 vs 19 of 2,778 paired measurements (95% CI 8.5–16.0 vs 2.7–4.3). University of Michigan cohort. Observational: race recorded; pigmentation not measured. It shows a device that looked accurate on average missing diagnosed hypoxaemia three times as often in one group. It does not justify a race-based correction.

Regulator response

US FDA draft guidance, 7 Jan 2025: clinical testing of pulse oximeters across a diverse range of skin tones, assessed on the Monk Skin Tone scale.

Who is missing from our data?

Ramesh - FoGO

- People using walking aids
- Cognitive fluctuation
- Crowded homes, floor-level living
- Sun or drafts over the sensor
- Public datasets with low or no Indian participants

Kamala - SwaKnee

- Women
- People with obesity
- Manual and agricultural workers
- People with other illnesses and medicines
- Rural users far from follow-up

NIHCT Center diversity guidance (2020): diversity includes comorbidities, concurrent medicines and environment; report subgroups. Stanford BioDesign health equity: equity is decided at identify, invent and implement.

Take two answers for each patient. Who would our data miss?

Sources: Spindig et al., NEJM 2020; US FDA pulse-oximeter draft guidance, 7 Jan 2025; NIHCT Center diversity guidance (2020); Stanford BioDesign health equity

19

SLIDE 19 • NEED AND EVIDENCE • 23:30–24:30 • 1.0 MIN • DEEP-DIVE (75-MIN PATH)

## Subgroup performance matters

Visual Native bar chart (Sjoding 11.7% vs 3.6%) + 'who is missing' cards for Ramesh and Kamala + FDA 2025 response pill

Interaction Who is missing from our data?

02

PART 2 • PEOPLE

Can they say no, and who gains if they say yes?

Consent that is free and understood, and interests that are declared and managed: the people side of translation, under Indian rules.

Decision 2 of 5: Ramesh's neurologist is also the inventor

What valid consent requires in India: ICMR 2017, NDCT 2019, Helsinki 2024

Conflicts of interest: disclose, then manage; the rules in India and abroad

WHAT A PARTICIPANT WOULD ASK

"Can I freely say no?"  
Helsinki 2024, para 27: free and informed relationship; consent must be sought by someone independent of it

"Do I understand what I am agreeing to?"  
ICMR 2017 s. 5–6: language the participant understands; an impartial witness if they cannot read

"Who gains if I say yes?"  
ICMR 2017: freedom, disclosure is the ethics conversation; ICM 2003 ch. 4.8 and ICMAPWD 2014 for doctors and companies

SLIDE 20 • PEOPLE • 24:30–24:45 • 0.25 MIN • DEEP-DIVE (75-MIN PATH)

## Can they say no, and who gains if they say yes?

Visual sectionDivider (no photo; three items)

Interaction —

WOULD YOU PROCEED? 2/5

His neurologist is the inventor

A fictional vignette built on real consent rules; it describes neither project's actual procedure

1

"Will saying no change my care?"

2

Consent form: English only

3

Only neurologist within 100 km

4

Twist, if time allows: the clinic also wants to film him for a talk. Does that change your vote?

5

Vote A, B or C. Then 60 seconds of role-play: Ramesh, the recruiting neurologist, an observer. Observer: what would make refusal easier?

Three options

A

Recruit as planned

The neurologist takes consent himself, this week

B

Proceed with safeguards

Which ones? Name two

C

Not yet

Fix the consent process before anyone is approached

Independent consent taker

Outs or Hindi, teach-back

Interest disclosed to the EC

Sources: Declaration of Helsinki 2024, para 27; ICMR 2017, ss. 5-6; CDMS 2016, Guideline 25; 45 CFR 46.116 (comparator); GMC recordings guidance (comparator)

21

SLIDE 21 • PEOPLE • 24:45–27:45 • 3.0 MIN • **CORE (40-MIN PATH)**

# Independent consent, local language, no exclusion

Visual factRows + optionCards

Interaction Vote; 60-second role-play

PEOPLE • CONSENT IN INDIA

What valid consent requires in India

Eight requirements of valid consent - ICMR 2017 ss. 5-7 - NDCT Rules 2019

1

Information, key facts first

Purpose, procedures, risks, benefits, alternatives; the US Common Rule (45 CFR 46.116) starts with key information

2

A language the participant understands

Outs or Hindi, not English only; plain words; teach-back (MBCT plain-language glossary)

3

An impartial witness if the participant cannot read

A literate witness unconnected to the research signs with the participant (ICMR 2017, s. 5)

4

Legally authorised representative where capacity is impaired

Fluctuating cognition, older and dependent people; LAR consent plus assent (ICMR 2017, s. 6)

5

Audio-visual recording where the rules require it

NDCT Rules: vulnerable participants in new drug trials; ICMR allows AV documentation, witness in frame

6

Re-consent when anything material changes

New risks, a new software version or protocol, ethics committee amendment, then the participant again

7

Withdrawal at any time, without penalty

Refusal or withdrawal must not change care (ICMR 2017; CDMS 2016 Guideline 19)

8

Compensation for research-related injury

Free medical management and compensation; insurance is on the CDSCO Form MD-22 checklist

Before the first participant is approached

A

Ethics committee registered with CDSCO - NDCT Rules 2019, Rules 7-8; HMR-2017 refers to the ethics committee in Rule 50

B

CIA permission, Form MD-22 → MD-23 - Seventh Schedule: consent form, investigator plan, insurance, data within one year

C

CTRI registration - prospectively, before the first participant is enrolled; pre-specified outcomes

Three special cases

A

Dependent relationship - Helsinki 2024, para 27

Consent must be sought by an appropriately qualified person independent of that relationship; disclose the investor's interest (CDMS 2016, Guideline 25)

B

Vulnerable participants - ICMR 2017, s. 6

Fluctuating cognition, old age, poverty or dependence on the clinic; extra ethics committee scrutiny, LAR consent with assent, caregivers involved

C

Filming and recordings - separate consent

Specific, reversible, in writing for public media; a gift video identifies a person; never a condition of care or of study entry (UK GMC as comparator)

Sources: ICMR 2017, ss. 2, 5, 6, 7; NDCT Rules 2019, Rules 7-8; HMR-2017 Chapter VII, Forms MD-22/23; CDSCO MD-22 checklist; Helsinki 2024, para 27; CDMS 2016, Guideline 19; 45 CFR 46.116 (comparator)

22

SLIDE 22 • PEOPLE • 27:45–29:00 • 1.25 MIN • **DEEP-DIVE (75-MIN PATH)**

# ICMR 2017, NDCT 2019 and Helsinki 2024 set the floor

Visual stepsVertical checklist (information, language, witness, LAR, AV recording, re-consent, withdrawal, compensation) + cardGrid of special cases (dependent relationship, vulnerable groups, filming) + ethics committee registration box

Interaction —

CONFLICTS OF INTEREST

Disclose, then manage

Interests

A

Equity, royalties or salary from the company that will sell the device

B

Grants and institutional reputation: £25.5 lakh received; a BRAC BIG proposal under review

C

Personal reputation as inventor, clinician and principal investigator

Risks

A

Pressure to enrol dependent patients; Ramesh and his only neurologist

B

Outsiders reading of outcomes; participants assume research is care chosen for them

C

Selective reporting; industry-sponsored drug and device studies reach favourable conclusions more often, SS 1.36 (January 2017)

Safeguards

A

Independent consent taker and independent outcome assessor

B

Prospective CTRI registration with pre-specified outcomes

C

Publish every result, null and negative included; plain-language summary to participants (NDCT 2017)

THE RULES, IN INDIA AND ABROAD

India

UCMPMD 2024 (amended 30 Apr 2024)

Voluntary code for device companies (DAP circular, 6 Sep 2024); no gifts, travel or hospitality for health professionals; claims accurate and capable of substantiation; complaints to UCMPMD within 90 days

USA

Open Payments (Sunshine Act)

Manufacturers of drugs and devices report consulting fees, travel, research payments, royalties and ownership interests to physicians every year. CMS publishes them

Abroad

USA - Harvard Medical School policy, 2010

From disclosure to beneficiaries: faculty named from industry speakers' bureaus, the model for 'disclose, then manage' in this session

UK

Cumberlege Review, 2020

First Do-Not-Harm recommended an expanded GMC register of all doctors' financial and non-pecuniary interests, a Patient Safety Commissioner and device registers

None of these replace project-level management: independent consent and assessment, a registered protocol, full publication

ICMR 2017 - NDCT Rules 2019

Interests disclosed to the ethics committee and in the consent conversation; an ethics committee registered with CDSCO reviews the investigator plan

Sources: Lumish et al., Casterline 2017; UCMPMD 2024 (amended 30 Apr 2024); IMC 2002, cl. 6.8; ICMR 2017; CMS Open Payments; HMS policy 2010; Cumberlege Review 2020

23

SLIDE 23 • PEOPLE • 29:00–30:00 • 1.0 MIN • **CORE (40-MIN PATH)**

# Interests → risks → safeguards; the rules in India and abroad

Visual Three-column flow (interests, risks, safeguards) + a rules strip: India (UCMPMD 2024, IMC 2002 cl. 6.8, ICMR 2017) vs abroad (US Open Payments, HMS 2010, UK Cumberlege)

Interaction —

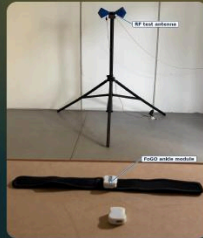
03

PART 3 • PERMISSION IN INDIA

## Who allows what, for which device?

One Act, one set of Rules, four risk classes. How a wearable like FoGO moves from a test licence to a State manufacturing licence, and what never stops afterwards.

- Four risk classes and the First Schedule rules that decide them
- Which licence, from which authority, with which form and fee
- What a licence holder must keep doing: PSURs, MvPI reports, change control



FoGO wearable module under IEC 60601-1-2 EMC pre-compliance testing, July 2024

SLIDE 24 • PERMISSION: REGULATION IN INDIA • 30:00–30:15 • 0.25 MIN • DEEP-DIVE (75-MIN PATH)

## Who allows what, for which device?

Visual sectionDivider with FoGO EMC test photo

Interaction —

INDIA • THE SYSTEM

### India's regulatory architecture

Drugs and Cosmetics Act, 1940 • s.3(b): a medical device is regulated as a 'drug' • all devices notified from 1 Apr 2020 (S.O. 648(E))

Medical Devices Rules, 2017 • G.S.R. 78(E), 31 Jan 2017 • In force 1 Jan 2018 • Rule 4: Class A, B, C, D by the First Schedule

#### Central Licensing Authority

CDSCO, headed by the DGCI, New Delhi; Medical Device and Diagnostics Division

- Manufacture Class C and D: MD-7 → MD-9
- Import, every class: MD-14 → MD-15
- Test licences MD-2 → MD-13, critical investigation MD-22 → MD-23
- No predicate in India: MD-26 → MD-27 (Rule 63); Medical Device Officers inspect C/D sites

#### State Licensing Authority

State or UT Drugs Controller, one per State

- Manufacture Class A and B: MD-3 → MD-5 (lean licence MD-4 → MD-6)
- Relies on a registered notified body's QMS audit of the site (Rule 12)
- Sale, stocking, distribution: MD-11 → MD-12, Rule 67A, since 30 Sep 2022
- Class A non-sterile, non-measuring: online registration instead of a licence (G.S.R. 777(E), 2022)

#### WHO ELSE IS IN THE ROOM



Sources: DCA Act 1940 s.3(b); MDR 2017 Rules 4, 13, 19, 67A; S.O. 648(E) 11 Feb 2020; CDSCO FAQ 2018; IPC MvPI; BIS IS 23485; NPPA/DPMD 2013

25

SLIDE 25 • PERMISSION: REGULATION IN INDIA • 30:15–31:45 • 1.5 MIN • DEEP-DIVE (75-MIN PATH)

## One Act, one set of Rules, many actors

Visual Org infographic: D&C Act 1940 → MDR-2017 → CDSCO/DCGI (Central Licensing Authority) and State Licensing Authorities; notified bodies; testing labs; MvPI/IPC; BIS; NPPA; ethics committees; CTIR — cards with who-does-what

Interaction —

INDIA • HOW WE GOT HERE

### From a handful of notified devices to all of them



#### 2024 • THREE DOCUMENTS TO READ BEFORE YOU FILE



Sources: MDR 2017 (23 S.R. 78(E)); S.O. 648(E) and G.S.R. 103(E) 11 Feb 2020; G.S.R. 743(E) 1777(E) 2022; CDSCO order 12 Oct 2022; Cabinet 26 Apr 2023; CDSCO MDSW guidance 21 Jul 2024; G.S.R. 743(E), 744(E) 14 Aug 2024; draft G.S.R. 515(E)

26

SLIDE 26 • PERMISSION: REGULATION IN INDIA • 31:45–32:45 • 1.0 MIN • DEEP-DIVE (75-MIN PATH)

## Regulation arrived in phases; the last phase closed in 2023

Visual timeline 1940 → 2005 → 2017 → 2018 → Feb/Apr 2020 → Oct 2021 → Oct 2022 → Oct 2023 → 2023 NMDP → 2026 MDSW guidance / proposed amendments

Interaction —

INDIA • RISK CLASSES

Four risk classes decide the route

Class A

Low risk

- Dressings, bandages
- Non-sterile exam gloves
- Clinical thermometer
- Deep-tissue TMS stimulator (CDSCO rehab list)

State - MD-5 (INDIAN regulator only)

Class B

Low-moderate risk

- Hypodermic needles, syringes
- Dr. needles (non-invasive)
- Hearing aids
- TENS systems (CDSCO list)
- FoGO: Class B (proposed)

State - MD-3 → MD-5

Class C

Moderate-high risk

- Inhalation pumps
- X-ray MRI systems
- Long-term infusion pumps
- Haemodialysis machines
- Orthopedic implants (joint, C or D, verify)

Central - MD-7 → MD-9

Class D

High risk

- Heart valves
- Implantable pacemakers
- Coronary drug-eluting stents
- Intra-uterine devices
- Contact with CNS or central circulation

Central - MD-7 → MD-9

FIRST SCHEDULE, PART I: FOUR RULE FAMILIES

1 Non-invasive

Intact skin only → A; channels or stores blood or body fluids → B

2 Invasive

Body orifice or surgical; transient, short-term; long-term; implants → C or D

3 Active

Therapeutic energy → B; hazardous energy or monitoring vital processes → C

4 Special rules

Medicinal substance → D; non-viable animal tissue → D; IVDs under Part II

WHERE OUR TWO DEVICES WOULD SIT

Class B (proposed)

FoGO: active wearable plus app that cares. The developer's proposal, not a CDSCO classification

Swaknee: CDSCO's rehabilitation list puts deep-tissue electromagnetic stimulation in Class A. The active-therapeutic rule suggests B.

Increasing risk → more evidence, higher authority, closer scrutiny

Sources: MDR 2017 Rule 4 and First Schedule; CDSCO classification notices 3 Sep 2020; Class A list 2025, merged list 13 Oct 2025; CDSCO rehabilitation list (via Morulus); consultancy examples (Registrars, Co-respond, Agile Regulatory); FoGO project material

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SLIDE 27 • PERMISSION: REGULATION IN INDIA • 32:45–34:45 • 2.0 MIN • **CORE (40-MIN PATH)**

# Class A to D by intended use, invasiveness, duration and energy

Visual classLadder A–D with verified examples and licensing authority + cardGrid of First Schedule rule families + where FoGO and SwaKnee would likely sit (labelled 'proposed')

Interaction —

INDIA • WHICH LICENCE

Which licence, from which authority, with which form

| Activity                                               | Class A - non-sterile, non-measuring                                                     | Class A sterile or measuring - Class B | Class C - Class D             | Who decides                           |
|--------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------|-------------------------------|---------------------------------------|
| Manufacture                                            | Online registration + Essential Principles self-certification (G.S.R. 777E), 14 Oct 2022 | MD-3 → MD-5                            | MD-7 → MD-9                   | State (A/B) - Central (C/D)           |
| Loan licence                                           | —                                                                                        | MD-4 → MD-6                            | MD-8 → MD-10                  | State (A/B) - Central (C/D)           |
| Import                                                 | MD-14 → MD-15                                                                            | MD-14 → MD-15                          | MD-14 → MD-15                 | Central, every class                  |
| Test licence: make or import for study, demo, training | MD-12 → MD-13 - MD-16 → MD-17                                                            | MD-12 → MD-13 - MD-16 → MD-17          | MD-12 → MD-13 - MD-16 → MD-17 | Central                               |
| Clinical investigation                                 | MD-22 → MD-23                                                                            | MD-22 → MD-23                          | MD-22 → MD-23                 | Central + registered ethics committee |
| No predicate in India (Rule 63)                        | MD-26 → MD-27                                                                            | MD-26 → MD-27                          | MD-26 → MD-27                 | Central                               |
| Sale, stocking, distribution                           | MD-41 → MD-42 (Rule 87A)                                                                 | MD-41 → MD-42 (Rule 87A)               | MD-41 → MD-42 (Rule 87A)      | State                                 |

A/B licence, 15,000 sites + 1000 device

C/D licence: ₹50,000 sites + ₹1,000 device

Import: US\$1,000 - 3,000 sites + \$10 - 1,000 device

MD-13: 1000 device

MD-41/42: ₹5,000

Total = State Licensing Authority - Indigo + Central (CDSCO). Licences are perpetual: pay the retention fee (same amount) every 5 years; late fee 2% a month; a manufacturing licence lapses after 180 days unpaid. Clinical investigation fee: see the Second Schedule (verify).

Sources: MDR 2017 and CDSCO regulatory pathway; Forms MD-3/5, 7/9, 12/13, 14/15, 22/23, 26/27, 41/42; Second Schedule fees (CDSCO fee help file); G.S.R. 777E and 754E 2022; FoGO project material

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SLIDE 28 • PERMISSION: REGULATION IN INDIA • 34:45–36:45 • 2.0 MIN • **CORE (40-MIN PATH)**

# The form depends on the class and the activity

Visual matrix: activity × class → form pairs and authority (manufacture, loan, import, test, clinical investigation, sale), colour-coded State vs Central; fee pills

Interaction —

INDIA • GETTING A LICENCE

The route to a manufacturing licence

1 - Build the QMS

Fifth Schedule ISO 13485 aligned. BIS IS 23485 handles QMS and Essential Principles. A certificate is not a legal precondition, but auditors expect alignment. Start-ups: 3–6 months, ₹5–8 lakh (consultancy estimate).

₹5,000 + ₹500

Class A/B licence fee

per site + per device; the same again at each 5-year retention

2 - Apply online

cdscodonline.gov.in Form MD-3 (Class A/B, to the State) or MD-7 (Class C/D, to CDSCO) with site and device details and the Essential Principles checklist.

₹50,000 + ₹1,000

Class C/D licence fee

per site + per device; CDSCO inspection, not a notified-body audit

3 - Audit or inspection

A: licence on documents within 45 days; notified-body audit within 220 days after B; notified-body audit, within 90 days, before grant. C/D: CDSCO Medical Device Officers inspect within 60 days (Rule 231B).

4–5 months

Typical MD-5 journey

consultancy estimate including the notified-body audit

4 - Licence

MD-5 (State, A/B) or MD-9 (Central, C/D). C/D decision within 45 days of the inspection report (Rule 214E). Perpetual; no renewal application.

140 → 115 days

Class B cap, draft 2026

G.S.R. 115E1, 23 Jan 2026; C/D 105 → 90 days; not final

5 - Keep it

Retention fee every 5 years (same amount); 2% a month late fee, deemed cancelled after 180 days unpaid. Rule 43A: suspension or cancellation after three-cause.

Class A is granted on self-certification and audited after; Class B is audited before grant; Class C and D are inspected by CDSCO. Higher class, closer scrutiny.

Sources: MDR 2017 Rules 22, 23, 43A and Fifth Schedule; BIS IS 23485; CDSCO fee help file; draft G.S.R. 115E1 23 Jan 2026; CDSCO retention fee notices; Standing Committee on Health 17900 report (24 Mar 2020); consultancy estimates (Co-respond, Inouga)

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SLIDE 29 • PERMISSION: REGULATION IN INDIA • 36:45–38:15 • 1.5 MIN • **DEEP-DIVE (75-MIN PATH)**

# QMS, audit or inspection, licence; then keep it

Visual chevronFlow (QMS per Fifth Schedule → application MD-3/MD-7 on the portal → notified-body audit (A/B) or CDSCO inspection (C/D) → licence MD-5/MD-9 → retention) with timelines and fees under each step + statTiles (fees per site/device by class; validity; retention)

Interaction —

INDIA • IMPORT AND REGISTRATION

### Importing, registering and relying on foreign approvals

**Import: Central Licensing Authority, every class**  
Rule 34-38, MDR-2017 / Form MD-14 → Science MD-14

- 1. Appoint an Indian authorised agent**  
The agent or importer files on the Medical Device online system (eSubordinating.gov.in)
- 2. Apply in Form MD-14 to CDSCO**  
One route for Class A, no; the overseas site may be inspected
- 3. Pay Second Schedule fees in US dollars**  
A: \$1,000/site + \$50/device • B: \$2,000 + \$1,000 • C-D: \$3,000 + \$1,500
- 4. Decision within 9 months**  
In parentheses queue 6-9 months; MD-15 is perpetual with a 5-yearly retention fee (90-day grace)
- 5. Report what happens abroad**  
Rule 38: withdrawal, restriction or cancellation elsewhere → CDSCO within 15 days. Test imports: MD-16 → MD-17, \$300/device

**Class A non-sterile, non-measuring; register, don't license**  
G.S.R. 777(8), 14 Oct 2022 (South Amendment)

- Upload on the online system: site name and address; device details: an undertaking that the device is Class A, non-sterile, non-measuring; a self-certified Essential Principles checklist
- Registration number goes on the label, Reg. No., wording proposed in the Dec 2023 draft
- QMS self-certification reviewed by G.S.R. 744(1), Aug 2026. Sterile or measuring Class A still needs MD-5

**Reliance: no predicate in India, approved abroad**  
Rule 43, Chapter III: Form MD-20 → permission MD-27

- CDSCO may waive the local clinical investigation if the device is approved in the US, UK, Australia, Canada, Japan or, since 14 Aug 2026 (G.S.R. 744(3)), the EU
- Conditions marketed there for at least 2 years; CDSCO satisfied with safety, performance and post-market data
- A post-marketing clinical investigation in India is still required

recognised regulators:

About 70% of what India uses is imported (Parliamentary Standing Committee, Mar 2024). Every importer is a licence holder with the same post-market duties as a manufacturer.

Sources: MDR-2017 Rules 34-38, 63 and Second Schedule; Form MD-14, G.S.R. 777(8) 14 Oct 2022; G.S.R. 744(1) 14 Aug 2026; Standing Committee on Health (Mar 2024); Medical Buyer FID4 Imports

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SLIDE 30 • PERMISSION: REGULATION IN INDIA • 38:15-39:15 • 1.0 MIN • DEEP-DIVE (75-MIN PATH)

## Imports are central; some foreign approvals shorten the evidence path

**Visual** Two panels: import route (MD-14 → MD-15, free sale certificate, authorised agent, fees) and Class A non-sterile non-measuring registration; plus a 'reliance' card for devices approved in listed countries

**Interaction** —

INDIA • WHAT A LICENCE HOLDER MUST DO

### Eight duties that never stop

**Essential Principles**  
Rule 4 - EP checklist

- Checklist filed with every application
- Class A non-sterile, non-measuring, self-certified

**Quality management system**  
Rule Schedule - QMS 234(8)

- ISO 13485-aligned manual, procedures, records
- Notified body audit (A/B) or CDSCO inspection (C/D)

**Labelling**  
Rule 44 - Fourth Schedule

- Name, manufacturer, net quantity, month and year of manufacture and expiry, licence number
- Form 14 Feb 2022: distribution site licence number (S.M.L.)

**Unique Device Identification**  
Rule 46 - data deferred

- Fixed start date removed by G.S.R. 938(1), 31 Dec 2021
- No date fixed in 2026: plan the UDI anyway

**Your licence**  
MD-5 or MD-9 - perpetual

- Retention fee every 5 years, same amount
- Rule 42A: suspension or cancellation after show-cause for non-compliance

**Standards**  
Rule 7 - BIS, then ISO/IEC

- BIS - 1,200 device standards, 254 critical devices prioritised
- Testing fees now in a Ninth Schedule (G.S.R. 743(8))

**Records and traceability**  
Chapter IIIA - Rule 47A

- Batch/lot serial number and software version for every unit
- Registration number on the label; others told MD-42

**Change control**  
Recent guidance, 21 Jul 2026

- Written control, validation, documented change management
- Major change prior approval, minor: notification (notify)

**Vigilance**  
Rule 48 - MDR January 13 May 2024

- Serious adverse events, report within 15 days
- Action abroad → CDSCO within 15 days, MDR form to MARI

Sources: MDR-2017 Rules 6, 7, 38, 43A, 44, 46 and Fifth Schedule; G.S.R. 743(8) 14 Aug 2026; G.S.R. 938(1) 31 Dec 2021; BIS 9 234(8); CDSCO MDRW guidance 21 Jul 2026; CDSCO circular 15 May 2024

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SLIDE 31 • PERMISSION: REGULATION IN INDIA • 39:15-40:30 • 1.25 MIN • DEEP-DIVE (75-MIN PATH)

## Essential principles, QMS, labelling, UDI, standards, records, change control, vigilance

**Visual** cardGrid 8 (icon, duty, rule/schedule, what to show) around the licence

**Interaction** —

INDIA • TESTING IN PEOPLE

### Clinical investigation under MDR-2017

**1. Study write**  
under MD-23, Seventh Schedule dossier

**2. Registered ethics committee**  
approves protocol, consent

**3. Form MD-22**  
received → permission MD-23

**4. Register before**  
the first participant

**5. Pilot → pivotal**  
MD5, compensation, final report

**Seventh Schedule dossier**  
Filed with MD-23 to CDSCO (Seventh)

- Clinical investigation plan
- Investigator's brochure
- Design control documents
- Design verification and validation report
- Essential Principles checklist
- Informed consent form
- Investigator undertaking
- Ethics committee approval
- Pilot, pivotal or post-marketing study?

**Exemptions and shortcuts**

- Rule 43 provides: approved in the US, UK, Australia, Canada, Japan or (since Aug 2026) the EU, marketed ≥ 2 years → local study may be waived; post-marketing study still due
- Academic study of a licensed device: ethics approval; data not for marketing (notify the Rule test)

Before the first participant: ethics approval, MD-23 permission, CTIR registration, insurance and compensation arrangements. Pilot = exploratory first-in-human; pivotal = confirmatory. Fee waived for government institutions (Rule 51(2)).

Sources: MDR-2017 Chapter VI, Rules 50-63 and Seventh Schedule; CDSCO Form MD-22 and checklist; NCTX Rules 2019; ICHM 2017; ISO 14155:2020; G.S.R. 744(8) 14 Aug 2026

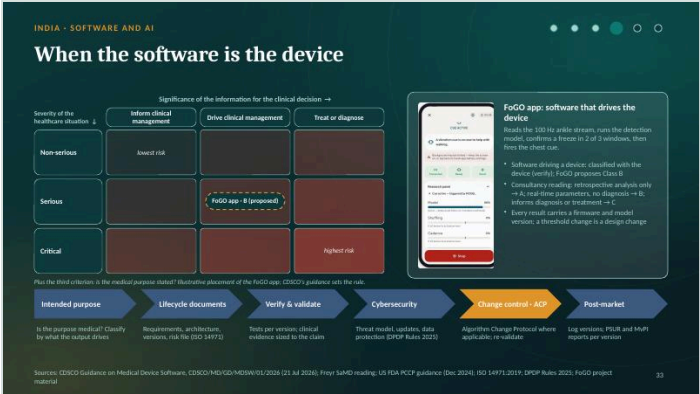
32

SLIDE 32 • PERMISSION: REGULATION IN INDIA • 40:30-42:00 • 1.5 MIN • DEEP-DIVE (75-MIN PATH)

## Pilot and pivotal; permission, ethics, registration, compensation

**Visual** Swimlane flow (sponsor / ethics committee / CDSCO / CTIR) + Seventh Schedule document checklist + exemptions (academic study; foreign-approved devices)

**Interaction** —

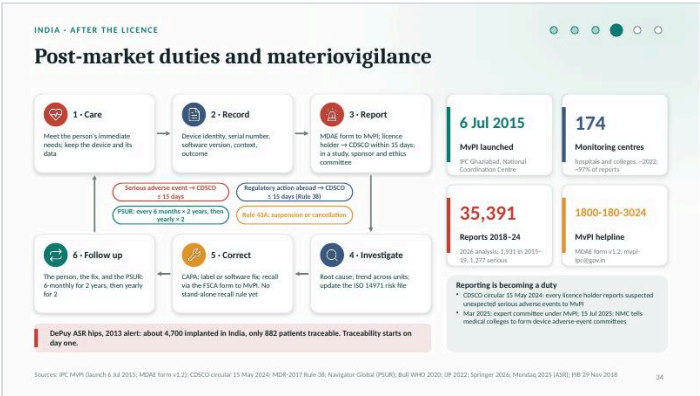


SLIDE 33 · PERMISSION: REGULATION IN INDIA · 42:00-43:15 · 1.25 MIN · DEEP-DIVE (75-MIN PATH)

## CDSCO's 2026 guidance: classify by what the output drives

Visual 2x2 classification grid (significance of information × healthcare situation) + lifecycle chevrons (documentation, verification, validation, change control, ACP) + FoGO app example

Interaction —

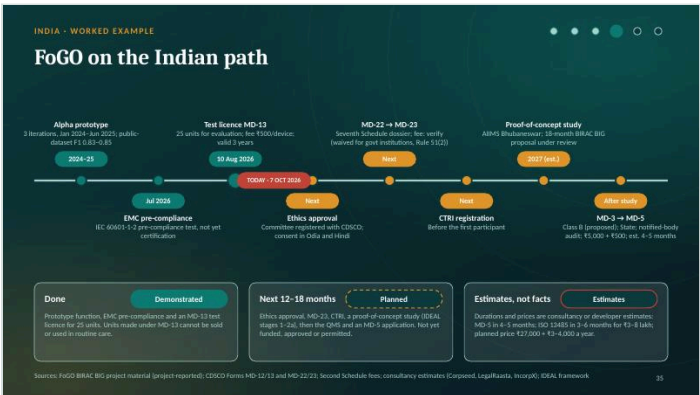


SLIDE 34 · PERMISSION: REGULATION IN INDIA · 43:15-44:30 · 1.25 MIN · DEEP-DIVE (75-MIN PATH)

## PSUR, adverse-event reporting, recalls and MvPI

Visual Loop infographic with timelines (PSUR cadence; reporting windows) + MvPI statTiles (launch, centres, reports, helpline) + recall/FSCA steps

Interaction —



SLIDE 35 · PERMISSION: REGULATION IN INDIA · 44:30-46:00 · 1.5 MIN · CORE (40-MIN PATH)

## Where a real Class B (proposed) wearable stands and what comes next

Visual timeline of done (EMC pre-compliance Jul 2026; MD-13 test licence 10 Aug 2026) and next (MD-22 → MD-23; ethics; CTIR; study; MD-3 → MD-5) with status chips, estimated durations and fees

Interaction —

WOULD YOU PROCEED? 3/5 • QUICK QUIZ

### Classify these four devices

**Digital thermometer**

- Measures body temperature at home
- Battery powered, a display, no app
- Notified as a "drug" from 3 Jan 2023
- NPPA trade-margin cap since Jul 2023

**PEMF knee device**

- Pulsed electromagnetic field to the knee
- 45-minute sessions at home (buffet)
- Applicator plus mains-powered controller
- Leaflet: adjust for knee osteoarthritis

**Wearable + app that cues**

- Axial sensor streams to a phone app
- App detects freezing of gait
- Chest module vibrates a cue
- Holds an MD-12 test licence for 25 units

**Coronary stent**

- Implanted in a coronary artery
- Stays in the body permanently
- Drug-eluting: NPPA ceiling price
- Notified before 2020; licensed since 2018

**THREE QUESTIONS THAT DECIDE IT**

Does it enter the body, and for how long? Inert skin, body, orifice, surgery, transient, short or long term.

Is it active? Energy delivered, software driving an action, monitoring of vital physiological processes.

What does the label claim? Intended use fixes the class, the authority and the evidence.

**A** Registration only if non-storable, non-measuring, also State MD-5

**B** State Licensing Authority - MD-3 → MD-5

**C** Central (CDSCO) - MD-7 → MD-9

**D** Central (CDSCO) - MD-7 → MD-9

**Tables, 90 seconds: write the class, the authority and the form for each device. Watch for the trap: a Class A device can still need a State licence.**

Sources: MDR 2017 First Schedule; CDSCO rehabilitation and merged classification lists; consultancy examples (Opson, RegisterKaro, Agile Regulatory, Corpseed); O.S.R. 777(2); NPPA steel prices (Apr 2026); PDCG project material

SLIDE 36 • PERMISSION: REGULATION IN INDIA • 46:00–48:00 • 2.0 MIN • **CORE (40-MIN PATH)**

## Apply the rules: class, authority, form

**Visual** Four device cards (digital thermometer; PEMF knee device; wearable with app that cues; coronary stent) with hidden answers revealed in notes; option badges A–D

**Interaction** Tables assign class and licence; reveal

INDIA • POLICY AND ECOSYSTEM

### Where India builds devices

- National Medical Devices Policy 2023**  
Cabinet, 26 Apr 2023; US\$ 1.1 bn (2023) → US\$ 50 bn by 2030; 16 strategies, incl. an export council
- PLI for medical devices - ₹3,420 crore**  
FY2020-21 to FY2027-28; 5% on incremental sales; by Mar 2025, 27 projects, ₹1,133 crore; 54 products
- Four medical device parks - ₹400 crore**  
Up to ₹100 crore each for shared infrastructure: Himachal, Tamil Nadu, Madhya Pradesh, Uttar Pradesh
- PRIP - ₹5,000 crore • Strengthening scheme - ₹500 crore**  
Industry-academia R&D; New 2024 scheme: clusters ₹110 cr, clinical studies ₹100 cr, skills ₹100 cr
- AMTZ Visakhapatnam - EPCMD (22 Sep 2022)**  
NABH, ENAC, electrical-safety and biomaterial lab; 3D printing; KIIT; export council at VEDA, Noida
- BIRAC BIG • MedTech Mitra • KIIT TBI**  
BIG: up to ₹50 lakh for 18 months; MedTech Mitra (ICMR • CDSCO) hand-holding; PuGO sits at KIIT TBI

**~70%**  
Imported  
Standing Committee, Mar 2024

**US\$ 8.18 bn**  
Imports, FY2023-24  
~60% electronic medical equipment

**US\$ 4.1 bn**  
Exports, FY2024-25  
From US\$ 2.3 bn in FY2020-21

Sources: Cabinet release 26 Apr 2023 (NMDP); Invest India and Indian Pharma Post (PLI); Business Standard 24 Sep 2023 (parks); Wikipedia and Budgetary PRIP (Nov 2024 scheme); AMTZ; BIRAC BIG; ICMR; MedTech Mitra; Standing Committee Mar 2024; Medical Device; IndiaBioscience

SLIDE 37 • PERMISSION: REGULATION IN INDIA • 48:00–49:15 • 1.25 MIN • **DEEP-DIVE (75-MIN PATH)**

## Policy, parks, PLI, test labs and grants

**Visual** mapPanel India (four device parks, AMTZ, KIIT/Bhubaneswar) + stepsVertical of schemes (NMDP 2023, PLI, parks, PRIP, EPCMD, BIRAC/DST) + import/export statTile

**Interaction** —

04

PART 4 • PERMISSION ABROAD

### Same questions, different answers

What is it for, how risky is it, what evidence, who keeps watching? Every regulator asks the same four questions and answers with different forms, fees and clocks.

**Who regulates where, and the forum that links them:** IMDRF, MDSAP, WHO

**Risk classes mapped across India, the USA, the EU, Japan, Canada and Australia**

**One dossier, many markets: the standards and reliance routes that travel**

PDCG table: medical, 2024; used device, classified repeatedly in every jurisdiction

SLIDE 38 • PERMISSION: REGULATION ABROAD • 49:15–49:30 • 0.25 MIN • **DEEP-DIVE (75-MIN PATH)**

## Same questions, different answers

**Visual** sectionDivider

**Interaction** —

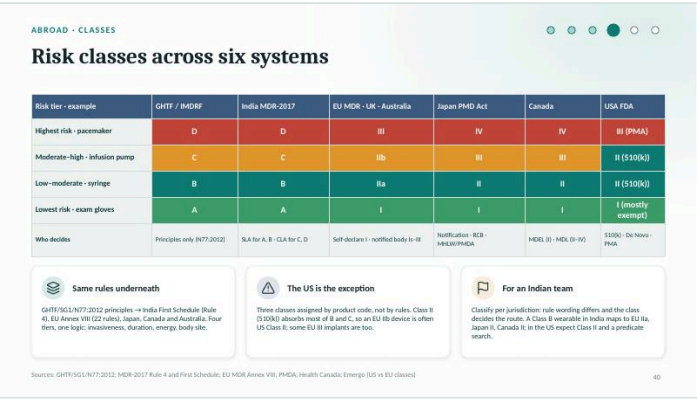


SLIDE 39 • PERMISSION: REGULATION ABROAD • 49:30-50:30 • 1.0 MIN • **CORE (40-MIN PATH)**

## Nine regulators, one forum

Visual mapPanel world with markers (CDSCO, FDA, EU notified bodies, MHRA, PMDA, NMPA, TGA, Health Canada, ANVISA, HSA) + cards (IMDRF, MDSAP, WHO framework)

Interaction —



SLIDE 40 • PERMISSION: REGULATION ABROAD • 50:30-51:45 • 1.25 MIN • **DEEP-DIVE (75-MIN PATH)**

## Four tiers almost everywhere; names differ

Visual matrix: India A/B/C/D vs GHTF A-D vs US I/II/III vs EU I/IIa/IIb/III vs Japan I-IV vs Australia/Canada; example device row (stent, BP monitor, thermometer)

Interaction —



SLIDE 41 • PERMISSION: REGULATION ABROAD • 51:45-53:00 • 1.25 MIN • **DEEP-DIVE (75-MIN PATH)**

## 510(k), De Novo, PMA; IDE for studies; QMSR from 2026

Visual chevronFlow of pathways with review goals, fees and annual counts + statTiles + what it means for an Indian exporter

Interaction —



ABROAD · WHAT TRAVELS

One dossier, many markets

ISO 13485:2016

Quality management. Accredited 31 Oct 2020, based on FDA, MHRA, MDCAP and India's PFR Schedule

ISO 14971:2019

Risk management, with ISO/TS 24971:2020 guidance. IEC 60601-1 Annex A referenced is updated

ISO 14155:2020

Clinical investigation GCP, published 23 Mar 2020, no transition. extensivex, CEC and DMC

ISO 10993-1

Biological evaluation inside the risk process, requires by context Type and duration

IEC 60601-1 ed. 1.2

Electrical safety and essential performance. (2005 + A1:2012 + A2:2020) IEC via 60601-1-2

IEC 62304

Software life cycle (2008 + A1:2015), updated 2 Aug Aug 2020. Two safety classes, AL/ML

IEC 62366-1

Usability engineering (2015 + A1:2020). Find use errors before patients do

IEC 81001-5-1:2021

Coversecurity life cycle (2020/development), EU harmonisation due 27 May 2020

Relevance routes an Indian designer can use

India · CDSCO

Medical devices (2017)

NDMA 2017 Notice of the CDSCO and use file

FDA · MDCAP provision

Its decisions count for TGA and MHRA reliance

CE certificate

accepted by TGA, valid for Great Britain to 2028.7.2020

MHRA (PMDA) · MHGAP

member, WHO Listed Authority, Aug. 2021

TGA · accept CE, TGA, Canada, Japan, MDCAP and HSA evidence

Health Canada · MDCAP

mandatory (D-FS), WHO Listed Authority

Lessons for Indian innovators: write the technical file open to ISO 13485 and ISO 14971; run the first study to ISO 14155:2020; choose the first foreign market by reliance: TGA and MHRA accept FDA and CE decisions, so one of those two unlocks three markets.

Sources: ISO 13485 review (Oct 2020); ISO 14971:2019; ISO 14155:2020; IEC 60601-1 A2:2020; IEC 62304 ed. 2 (Revised); IEC 81001-5-1; TGA and MHRA reliance; MHGAP; WHO WLA

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SLIDE 45 · PERMISSION: REGULATION ABROAD · 56:45-58:00 · 1.25 MIN · DEEP-DIVE (75-MIN PATH)

# Harmonised standards and reliance routes reduce rework

Visual Standards wheel/cards (ISO 13485, 14971, 14155, 10993; IEC 60601-1, 62304, 62366-1, 81001-5-1) + reliance arrows (India ↔ US/EU/Japan/Australia/Canada) + lessons for Indian innovators

Interaction —

05

PART 5 · ACCESS AND SAFETY

Can they afford it, use it, and who answers later?

Capability audits, the total cost of a course and honest brochures before launch; a named owner for every step after it.

Who can't use it? Seven capabilities, plus the demands India adds

Total cost of use, and how far NPRA's price powers reach under DPCO 2013

Decisions 4 and 5: a brochure, a software update; then the harm loop and MvPI

Swaknee tablet controller and knee applicator (supplied product photographs by patient shown)

SLIDE 46 · ACCESS AND SAFETY · 58:00-58:15 · 0.25 MIN · DEEP-DIVE (75-MIN PATH)

# Can they afford it, use it, and who answers later?

Visual sectionDivider with SwaKnee controller photo

Interaction —

ACCESS · INCLUSIVE DESIGN

Who can't use it? Audit the demands a device makes

Illustrative ratings of the demand each task makes on the seven capabilities of the Cambridge EDC Inclusive Design Toolkit

| Capability        | What FoGO asks of Ramesh                                           | Demand | What Swaknee asks of Kamala                                   | Demand |
|-------------------|--------------------------------------------------------------------|--------|---------------------------------------------------------------|--------|
| Vision            | Read the app screen; see the LED and the glowing prompts           | Medium | Read the controller display; set the session time             | Medium |
| Hearing           | App alerts only; the cue itself is a vibration                     | Low    | Bleeps and error tones from the controller                    | Low    |
| Thinking          | Pair the phone, keep the charging routine, trust a cue mid-release | High   | Keep a 45-minute daily schedule for 45 days; read error codes | Medium |
| Communication     | Report a fault; consent and instructions in Odia or Hindi          | Medium | Follow written steps; ask for help when the cuff slips        | Medium |
| Locomotion        | Walk and turn with an ankle module while freezing                  | High   | Travel to the clinic for fitting and follow-up                | Medium |
| Reach and stretch | Bend to the ankle; place the chest patch                           | High   | Send to wrap the applicator around the knee                   | High   |
| Dexterity         | Strap buckle; adhesive patch; a small charging port                | High   | Fasten the cuff; connect the cable to the controller          | Medium |

Pick one demand to redesign first. Classroom simulation: fasten the strap wearing thick gloves. FoGO's team reports consulting 30+ potential users; the common ask was comfort and easier use (project-reported).

Make it testable: IEC 62366-1 usability engineering turns each demand into a requirement. MHCT Accessibility by Design (2021) adds home visits, travel support and accessible consent formats.

India adds demands the toolkit does not list

Language

Odia, Hindi or English on screen, in the leaflet and at consent

Literacy

Pictorial steps and teach-back, not a printed manual

Power

Curt and voltage swings: charging routines fall first

Smartphone

Shared, old, or none; no data plan for updates

Caregiver

Fitting, charging and reporting fall on a relative

Exclusion Calculator

In percentages rest on UK survey data (1996-97); scores present them as Indian figures.

Sources: Cambridge EDC Inclusive Design Toolkit (seven capabilities); Exclusion Calculator UK data 1996-97; MHCT Accessibility by Design (2021); FoGO project material (user consultation, project reports)

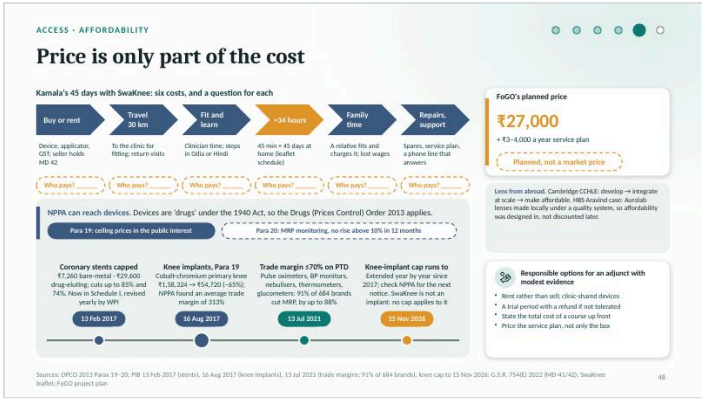
47

SLIDE 47 · ACCESS AND SAFETY · 58:15-59:15 · 1.0 MIN · DEEP-DIVE (75-MIN PATH)

# Audit the demands a device makes

Visual Capability audit table (seven Cambridge capabilities × FoGO/SwaKnee ratings) + India-adds panel + redesign prompt

Interaction Pick one demand to redesign

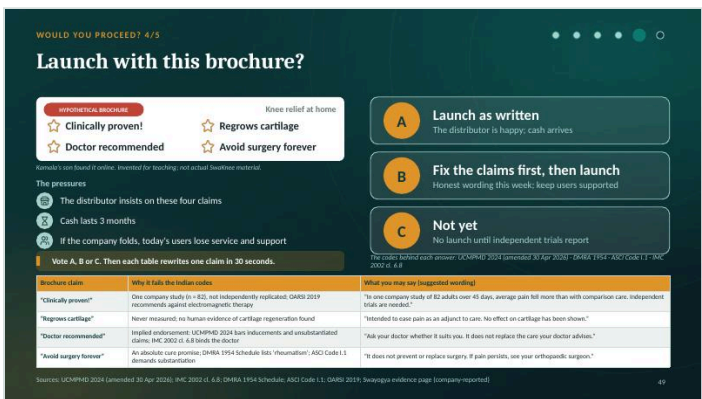


SLIDE 48 · ACCESS AND SAFETY · 59:15-60:45 · 1.5 MIN · **CORE (40-MIN PATH)**

## Total cost of use; NPPA's reach

Visual Cost journey with '?' slots + NPPA panel (DPCO 2013 powers; stents 2017; knee implants 2017; trade-margin cap 2021; 10% annual MRP rule) + FoGO planned price (Planned)

Interaction Who pays each step?



SLIDE 49 · ACCESS AND SAFETY · 60:45-62:45 · 2.0 MIN · **CORE (40-MIN PATH)**

## Claims must match evidence: UCMPPMD, DMRA, ASCI

Visual Brochure mock-up + pressures + optionCards + 'what you may say' mini-table

Interaction Vote; rewrite one claim



SLIDE 50 · ACCESS AND SAFETY · 62:45-64:15 · 1.5 MIN · **CORE (40-MIN PATH)**

## Evidence belongs to a version; changes need governance

Visual App screenshot + factRows + optionCards + change-control strip (CDSCO MDSW 2026; FDA PCCP)

Interaction Vote

**SAFETY**

## When harm happens, who answers?

Suppose Ramesh falls after a missed cue. Follow the report around the loop and write an owner under each step.

**1 - Care**

Meet his immediate needs, keep the device and its data, no basket! "stop using it"

**2 - Record**

Device identity, serial number, software version, context, outcome

**3 - Report**

MvPI MDAE form, license holder → CDSCO + 15 days, in a study → sponsor and ethics committee

**4 - Investigate**

Meet cases and context, trend across units; update the ISO 14973 risk file

**5 - Correct**

CAPI, labelling or software fix, recall through the PSCA form to MvPI

**6 - Follow up**

The person, did the fix work, PSUR 6 monthly for 2 years, then yearly for 2

Owner: \_\_\_\_\_

Owner: \_\_\_\_\_

Owner: \_\_\_\_\_

Owner: \_\_\_\_\_

Owner: \_\_\_\_\_

**India's reference case: DePuy ASR metal-on-metal hips, recalled worldwide in Aug 2010. CDSCO device alert Dec 2013. Import licence cancelled**

**≈4,700**  
ASR hips in India  
Surgeries 2004–2010; patients had to be traced through a helpline

**1,080**  
Traced by Aug 2018  
275 had revision surgery (Agarwal committee, 19 Feb 2018)

**₹20 lakh**  
Compensation base  
Formula approved 29 Nov 2018; awards ₹20 lakh for ₹1.25 crore by disability and age

Traceability starts on day one: a serial number on every unit, a registry, an in-app 'report a problem' button, a phone line that answers.

**Report to MvPI (IPC Ghaadabad, since 6 Jul 2015)**

MDAE form v1.2  
Open to patients, importers, distributors, clinicians and patients; PSCA form for recalls

1800 180 3024 - mvi-ipc@gov.in  
Helpline Mdn-PI 0140-5736; 174 monitoring centres (2022 figure)

CDSCO circular, 15 May 2024  
Every license holder must report suspected/unexpected serious adverse events to MvPI

Ahead: mostly passive too  
UK: EU: Agree and share info monthly on passive reporting (Ranier 2018), the UK Cambridge Review (2020) urged a device registry and a Patient Safety Commissioner

Sources: IPC MvPI (6 Jul 2015); MDAE form v1.2; CDSCO circular 15 May 2024; MvPI MvPI ASR committee 19 Feb 2018; PB 24 Nov 2018 (compensation formula); Ranier 2013; Cambridge 2020

SLIDE 51 • ACCESS AND SAFETY • 64:15–65:30 • 1.25 MIN • DEEP-DIVE (75-MIN PATH)

## Care, record, report, investigate, correct, follow up

Visual Loop with owners + ASR hip case statTiles (4,700 implanted; 1,080 traced; ₹20 lakh base) + MvPI reporting card

Interaction Assign an owner to each step

**CLOSING VOTE**

## Would you let a patient use it now?

The same three facts as the opening vote

A working prototype

Promising results on public datasets

A test licence granted

Four questions to answer silently before you vote

Which patient? Which use?

Which evidence? Who is responsible?

Opening: 1 = \_\_\_\_\_ 2 = \_\_\_\_\_ 3 = \_\_\_\_\_  
Now: 1 = \_\_\_\_\_ 2 = \_\_\_\_\_ 3 = \_\_\_\_\_

A patient-ready decision names

the patient the purpose the evidence the facts who answers

Vote again. Then two people whose vote changed: what changed your mind?

Sources: PICO project material (MD-13 test licence, 25 units, 10 Aug 2016; public dataset F1, project reports); MDR 2017 Forms MD-12/13 and MD-22/23; NCT: Rules 2019

SLIDE 52 • CLOSE • 65:30–67:00 • 1.5 MIN • CORE (40-MIN PATH)

## Same facts, same options; what changed?

Visual Same three facts + four silent questions + optionCards

Interaction Re-vote; compare

**TAKEAWAY • 90 SECONDS**

## Commit to one change in your own project

**COMMITMENT CARD** • one device, study or project you are working on

**Before** \_\_\_\_\_  
(start date)

**I will** \_\_\_\_\_  
(action)

**Owner:** \_\_\_\_\_  
(name)

**I proceed only if** \_\_\_\_\_  
(condition)

90 s: write 60 s: read it to a neighbour Two examples for the room

**Why a card:** Stanford BioDesign's Principled Decision-Making brief advises teams to write their ethical principles down early, before the pressured moment arrives. This exercise makes that advice personal; keep the card, or copy it into the printed checklist.

**Worked example**

"Before our first home test, I will register the study on CTBL. Owner: me. I proceed only if the ethics committee approves and an independent outcome assessor is named."

**One example commitment per stage**

- Need**  
Rewrite the need statement solution-neutral, screen it with ten users before inventing further ideas
- Evidence**  
No 'detect' claim until tested on data from Indian patients, with results by subgroup
- People**  
Name an independent consent-taker before anyone is approached; consent in Odia or Hindi
- Permission**  
Confirm the risk class and hold the MD-13 test licence before a unit leaves the lab
- Access**  
Cost the whole course for the household, not the device alone; decide rent or sell
- Safety**  
Serial number on every unit; 'report a problem' in the app; name who can pause a release

Sources: Stanford BioDesign, Principled Decision-Making (2022); CDSCO Form MD-12/MD-13 (test licence)

SLIDE 53 • CLOSE • 67:00–69:30 • 2.5 MIN • CORE (40-MIN PATH)

## Before [next step], I will [action]

Visual Commitment card + worked example + icon

Interaction Write, share

Patient-impact decision checklist

| Stage      | The patient asks                        | Evidence to see                                                                                                                                                                 | Permission in India (class - form - authority)                                                                                                                                                                    | Owner | Decision                                                        |
|------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------|
| Need       | "Will it help me?"                      | A solution-meets-need statement, intended use and claim in one sentence. The wording of the claim sets the evidence bar (Stanford Biologics)                                    | Is it a device? CDSC Act 1, 2019(a): every device regulated since 1 Apr 2020. Rule class A-5 by MDR-2017 Rule 4 and the First Schedule; check CDSCO's classification lists                                        |       | <input type="checkbox"/> Go<br><input type="checkbox"/> Not yet |
| Evidence   | "Is it proven, for people like me?"     | IDEA: 0 stage reached: results by subgroup (MCT 2020); an ISO 14177 risk file; a version log for software                                                                       | Prototypes: test licence MD-12 → MD-13 (CLA, ₹500 per device; 3 years); Study: MD-22 → MD-21 with Schedule documents; CTR before the first participant                                                            |       | <input type="checkbox"/> Go<br><input type="checkbox"/> Not yet |
| People     | "Can I say no? Who gave it to say yes?" | Consent in the participant's language; an independent consent taker where a patient depends on the researcher (Mehra 2024, para 27); informed disclosure and manager (DNR 2017) | Ethics committees registered with CDSCO (NCT Rule 2017; Rule 7-8; MDR Rule 5B); Companies: UCMPHD 2024; Doctors: IAC 2002 (1.6.9)                                                                                 |       | <input type="checkbox"/> Go<br><input type="checkbox"/> Not yet |
| Permission | "Is it allowed for this use?"           | QIR to the FPIB Schedule (SD 1488) aligned: Essential Principles checklist; Rule 4 (Labeling, software (the code) and change control (CDSCO MDSW guidance, 21 Jul 2020))        | Class A/B: MD-3 → MD-5, State Licensing Authority; notified-body audit; Class C/D: MD-7 → MD-9, CDSCO. Class A non-sterile non-measuring: registration only (G.S.R. 7776), 2022; import, any class: MD-14 → MD-15 |       | <input type="checkbox"/> Go<br><input type="checkbox"/> Not yet |
| Access     | "Can I afford it and use it?"           | A capability audit (Cambridge toolset) plus the Indian demand; the total cost of a course and who pays each step; claims matched to evidence                                    | NPM under DPCO 2013: Para 17 calling clients and loose implants; 2017; Para 20 MRP monitoring; Trade-marks cap (2021); Classes: UCMPHD 2024, DMRs 1954-ASCI Code; Salure: MD-41 → MD-42 (State)                   |       | <input type="checkbox"/> Go<br><input type="checkbox"/> Not yet |
| Safety     | "Who answers if it fails?"              | A named owner for each step of the harm loop: serial numbers and a registry; monitoring and a rollback trigger for every software change                                        | NAF-MDAE form (PC: 1800 180 3024); licence holder reports serious events to CDSCO within 15 days (Circular 13 May 2024); PMR 6-monthly + 2 years; then yearly + 2; Rule 53A suspension                            |       | <input type="checkbox"/> Go<br><input type="checkbox"/> Not yet |

ⓘ A workshop synthesis, not a substitute for ethics, scientific, regulatory or quality review. Every 'Not yet' needs an owner and a condition for proceeding. A licence, patent, grant or famous partner is not evidence of readiness.

Sources: Stanford Biologics; IDEAL 0; MCT 2020; ISO 14177; Hsai 2024; KMR 2017; NCT 2017; MDR-2017 (Rules 4, 44, 43A; Forms MD-5/a, 7/9, 12/13, 14/13, 22/23, 41/42); G.S.R. 7776); CDSCO MDSW 2020; DPCO 2013; UCMPHD; DMRs; ASCI; NAF.

SLIDE 54 • CLOSE • 69:30–70:00 • 0.5 MIN • **CORE (40-MIN PATH)**

Six stages, evidence to see, owner, go/not yet, Indian permission

Visual matrix with a 'Permission in India' column (class, form, authority)

Interaction —

MDR-2017 forms at a glance

State - SIA (trial)      Central - CDSCO, CLA (final)

| Activity                 | Class or scope                                                    | Apply to            | Granted as          | Who decides   | Rule, schedule or note                                                                                                          |
|--------------------------|-------------------------------------------------------------------|---------------------|---------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| Manufacture              | Class A, non-sterile and non-measuring                            | Online registration | Registration number | Online portal | G.S.R. 7776), 14 Oct 2022: self-certified Essential Principles checklist and undertaking; no licence                            |
| Manufacture              | Class A sterile or measuring; Class B                             | MD-3                | MD-5                | State (SIA)   | Rule 20; a Rule 13 notified body audits the site against the FPIB Schedule QMS                                                  |
| Manufacture              | Class C; Class D                                                  | MD-7                | MD-9                | Central (CLA) | Rules 21–23; CDSCO Medical Device Officers inspect the site before grant                                                        |
| Loan licence             | Class A; Class B                                                  | MD-4                | MD-6                | State (SIA)   | Same fee as the manufacturing licence (Second Schedule)                                                                         |
| Loan licence             | Class C; Class D                                                  | MD-8                | MD-10               | Central (CLA) | MD-10 issued by the CLA since G.S.R. 1889), 6 Mar 2019                                                                          |
| Test licence             | Make units for study, test, evaluation, demonstration or training | MD-12               | MD-13               | Central (CLA) | ₹500 per device. FOGO holds MD-13 for 25 units (30 Aug 2020)                                                                    |
| Test licence             | Import units for the same purposes                                | MD-16               | MD-17               | Central (CLA) | US\$100 per device                                                                                                              |
| Clinical investigation   | Investigational device, any class                                 | MD-22               | MD-23               | Central (CLA) | Chapter VII, Seventh Schedule dossier, including the ethics committee approval                                                  |
| New device, no predicate | Import on manufacture under Rule 63                               | MD-26               | MD-27               | Central (CLA) | Local study waivable if approved in the US, UK, Australia, Canada, Japan or EU (G.S.R. 7440), 14 Aug 2020) and marketed 2 years |
| Import                   | Any class, through an Indian authorised agent                     | MD-14               | MD-15               | Central (CLA) | Rules 34–36; decision within 9 months                                                                                           |
| Sale and distribution    | Any device, including IVDs                                        | MD-41               | MD-42               | State (SIA)   | Rule 87A (G.S.R. 7546), 30 Sep 2022; ₹3,000; deemed granted unless rejected in 30 days                                          |

Also named in the Rules: MD-2 and MD-40 are 'Certification of Registration' alongside MD-42 (G.S.R. 7436), 14 Aug 2020; MD-44, a test report form, is proposed in draft G.S.R. 8938), 4 Dec 2025; MDP performance evaluation forms are omitted because their numbers were not verified. Every application is filed on [oicdschedule.gov.in](#).

Sources: MDR-2017 and CDSCO regulatory pathway sheet; Forms MD-12, MD-14, MD-16, MD-22 ([Johns.gov/MD](#)); G.S.R. 1889), 6 Mar 2019; G.S.R. 7546), 30 Sep 2022; G.S.R. 7776), 14 Oct 2022; G.S.R. 7436) and 7440), 14 Aug 2020; FOGO BRCs presentation (MD-13, 30 Aug 2020).

SLIDE 55 • APPENDIX • REFERENCE • DEEP-DIVE (75-MIN PATH)

MDR-2017 forms at a glance

Visual matrix of application → grant form pairs by activity (MD-1..MD-46 as verified)

Interaction —

Fees and statutory clocks by class (Second Schedule)

| Second Schedule fee                                                  | Class A - low risk                                                                                                                                                                             | Class B - low-moderate                                                                | Class C - moderate-high                                                                                               | Class D - high                      |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Who licenses manufacture                                             | State Licensing Authority                                                                                                                                                                      | State Licensing Authority                                                             | Central Licensing Authority (CDSCO)                                                                                   | Central Licensing Authority (CDSCO) |
| Manufacturing or loan licence - MD-5/MD-4 (A, B) - MD-9/MD-10 (C, D) | ₹5,000/site + ₹500/device (non-sterile, non-measuring; registration only)                                                                                                                      | ₹5,000/site + ₹500/device                                                             | ₹50,000/site + ₹1,000/device                                                                                          | ₹50,000/site + ₹1,000/device        |
| Retention fee every 5 years (licence is perpetual)                   | ₹5,000/site + ₹500/device                                                                                                                                                                      | ₹5,000/site + ₹500/device                                                             | ₹50,000/site + ₹1,000/device                                                                                          | ₹50,000/site + ₹1,000/device        |
| Import licence - MD-13 (rules) - MD-17 (import)                      | US\$1,000/site + US\$1,000/device                                                                                                                                                              | US\$1,000/site + US\$1,500/device                                                     | US\$3,000/site + US\$1,500/device                                                                                     | US\$3,000/site + US\$1,500/device   |
| Test licence - MD-13 (rules) - MD-17 (import)                        | All classes: ₹500 per device to make (MD-13); US\$100 per device to import (MD-17)                                                                                                             |                                                                                       |                                                                                                                       |                                     |
| Sale registration - MD-42 (Rule 87A)                                 | All classes: ₹3,000, with a ₹1,000 retention fee                                                                                                                                               |                                                                                       |                                                                                                                       |                                     |
| Statutory clock                                                      | Class A                                                                                                                                                                                        | Class B                                                                               | Class C                                                                                                               | Class D                             |
| Manufacturing licence, current rules                                 | licence on documents within 45 days (no prior audit; notified body audit within 120 days after grant)                                                                                          | notified-body audit within 90 days, before grant; reported and re-audited to 140 days | Inspection within 60 days (Rule 23(1)); decision within 45 days of the report (Rule 21(4)); reported: 105 to 150 days | As Class C                          |
| Draft G.S.R. 5158), 23 Jun 2026 (not final on 7 Oct 2026)            | No change proposed                                                                                                                                                                             | 30 + 30 + 20 + 15 + 20 days: 140 → 115 days                                           | Rule 21(4) 45 → 30 days; Rule 23(1) 60 → 35 days; 105 → 90 days                                                       | As Class C                          |
| Import licence - MD-13; retention fee missed                         | Import; decision within 9 months of application (3 to 9 months in practice); Retention: late fee 7% a month; licence deemed cancelled after 300 days unpaid (manufactured) or 90 days (import) |                                                                                       |                                                                                                                       |                                     |

ⓘ Where the number(s) come from: Second Schedule, Medical Devices Rules 2017 (G.S.R. 786), 31 Jan 2017; G.S.R. 7436), 14 Aug 2020, adds a Ninth Schedule of testing fees under Rule 17 and 68 (Samples not captured); Draft G.S.R. 8838), 4 Dec 2025, restores partial validity for MD-2, MD-5, MD-6, MD-9, MD-10 and MD-13. Clinical investigation (MD-22) fee: verify on [oicdschedule.gov.in](#) before quoting.

Sources: Second Schedule; MDR-2017 (CDSCO portal fee file); Lawtext and Corporate fee pages; Applied Clinical Trials 2017; draft G.S.R. 5158), 23 Jun 2026 (G.S. Runa, Axis Actual); G.S.R. 7436), 14 Aug 2020; CDSCO retention fee notice (Suppression 2024)

SLIDE 56 • APPENDIX • REFERENCE • DEEP-DIVE (75-MIN PATH)

Fees and timelines by class

Visual matrix of fees (manufacture, loan, import, test, clinical investigation, retention) and statutory timelines

Interaction —

CDSCO documents to keep at hand

| Document                                        | Number or file                                                                                                               | Date                             | What it covers                                                                                                                                                                        | Status       |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Medical Devices Rules, 2017 (consolidated text) | G.S.R. 761(1), Feb. 2017 PDF on cdso.gov.in                                                                                  | 01 Jan 2017, in force 1 Jan 2018 | Classes A-D (Class A, First Schedule); licenses by class; PMH Schedule QMS; Rule 61 labelling; Chapter VII clinical investigation; Rule 61 devices without a predicate                | In force     |
| FAQ on Medical Devices Rules, 2017              | CDSCO/FAQ/MDO/01/2018; IVD FAQ QP-2022; addendums                                                                            | 2018, Mar 2022                   | Licensing in plain language; Rule 13 notified bodies (NABCB-accredited) audit Class A and B sites; IVD examples; specimen tubes Class A, alcohol analyzer Class B                     | Final        |
| Regulatory pathway under MDH-2017               | RegulatoryMSH-2017.pdf                                                                                                       | —                                | One-sheet map of forms and authorities; MD-3/MD-5 and MD-7/MD-9 manufacture; MD-16/MD-15 import; MD-12/MD-13 test licence                                                             | Overview     |
| Device classification lists                     | Public notices of 3 Sep 2020; category-wise lists                                                                            | 3 Sep 2020 onward                | About 1,866 devices in 24 categories (Class B alone 779) and 80 IVDs in 3, later: parafrase: 27 Aug 2023; Cardiovascular medical (Apr 2023), Interventional radiology (187)           | Living lists |
| Checklist for Forms MD-22                       | ChecklistFormMD-22MD.pdf                                                                                                     | —                                | Seventh Schedule dossier: investigation plan, investigator's brochure, design V&V, Essential Principles Checklist, consent form, ethics approval, clinical, pivotal or post-marketing | Checklist    |
| Guidance on Medical Device Software             | CDSCO/MD/CD-MDSW/01/2024                                                                                                     | 21 Jul 2024 (draft 21 Oct 2023)  | 62 pages (a4m); software in devices, stand-alone and networked systems, AI/ML, IVD software; class by intended use; lifecycle, cybersecurity, AI change planning, post-market duties  | Final        |
| Guidance on post-market surveillance            | Draft Guidance on PMS (2018); CDSCO/IVD/CD/PMH/01/2022                                                                       | 2018; 7 Jul 2022                 | PMH expectations for licence holders; the IVD draft targets Class C and D and point of care or home-use IVDs; neither reported as finalised                                           | Draft        |
| Matovigilance circular and forms                | Circular of 15 May 2024; MvM MD&E form v1.2; IPC draft IVD-MD-AE form revd; See.pdf (seecompliance.gov.in); gaestte 2237.pdf | 15 May 2024, 17 Feb 2025         | Every licence holder reports suspected serious adverse events to MvM (IPC); helpline 1800-180-9024; PSCA form for recalls and corrections                                             | Circular     |
| Free help file, testing laboratories            |                                                                                                                              | 2017, 2018                       | Several Schedule Free by class; Central Medical Device Testing Laboratories: NIB Noida (INDO), CDTL Chennai, CDTL Kolkata, RDTL Coimbatore, CDTL Mumbai                               | Reference    |

Amendments to know, by gazette number

|                                                                                              |                                                                                          |                                                                                          |                                                                                        |                                                                                                         |                                                                                                              |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| G.S.R. 1020(1)-11 Feb 2020<br>All devices regulated from 1 Aug 2020; Chapter 8A registration | G.S.R. 7400 - 30 Sep 2022<br>Rule 67A; MD-41 → MD-43 for sale, stocking and distribution | G.S.R. 7770(1)-14 Oct 2022<br>Class A non-durable non-measuring registration, no licence | G.S.R. 7431(1)-14 Aug 2024<br>Rule 44(4) distribution site label from 14 Feb 2022; MvM | G.S.R. 7400(1)-14 Aug 2024<br>P2 approvals added to the Rule 61 website; Class A QMS self-certification | Orally: 4 Dec 2023 - 23 Jan 2024<br>G.S.R. 853(1): perpetual validity; MD-44, G.S.R. 1110(1): license checks |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|

Sources: CDSCO documents as numbered (FAQ 2018; MD-22 checklist; MDSW guidance 21 Jul 2024; draft PMH 2018; IVD PMH 7 Jul 2022; circular 15 May 2024; for Rec; CMTL; gaestte); classification notices 3 Sep 2020 (PmM); G.S.R. 1020(1), 7540(1), 7540(1), 7710(1), 7430(1), 7440(1); drafts 8030(1) and 9101(1)

CDSCO guidance documents to keep at hand

Visual matrix of guidance documents with number, date and what it covers  
Interaction —

Evidence to request before stronger claims

| Question        | FoGO: freezing-of-gait wearable (Class B, proposed)                                                                                                                                                  | Swaknee: P250P knee system                                                                                                                                              | Benchmark or standard                                                                                                                                    |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Outcome         | Freeze detection against video-analysed events; then falls, confidence and walking outcomes, not only an F1 score                                                                                    | Pain and function against patient-important thresholds; structural (cartilage) outcomes only if structural claims are made                                              | Highest benchmark: 31 patients, 127 video-labelled events; Laboratory (2015); Knee-QA pain index: 18.9 mm, -46.8% (Dobson 2005)                          |
| Comparison      | Participant-level validation on unseen users; cue versus no-cue or sham cue                                                                                                                          | Randomised, blinded, sham-controlled; prospectively registered (CTR); cue interventions recorded                                                                        | RESCUE trial; n = 153; home curing; modest gait gains, falls not counted (2007); Cochrane 2023: 9 trials, 459 adults; pain about 15/100 better than sham |
| Who and where   | Diverse stage, walking aids; homes, turns and doorways, not only the laboratory                                                                                                                      | Age, BMI, severity, comorbidities; rural access and supervision                                                                                                         | Vitalable reviews; sensitivity 75–100%, specificity 45–100%; mostly Laboratory (2017); Knee-QA 28.7% in a five-site Indian community survey (2016)       |
| Burden and harm | False cues per hour and latency in real use; missed events, signal loss, skin data                                                                                                                   | Adverse events; adherence to 45-minute daily sessions over 45 days; support needs; durability after the course                                                          | ISO 14971:2019 risk management; IEC 62366-1; usability engineering; ISO 14155:2020 good clinical practice for device studies                             |
| Version         | Firmware and model version stated for every result                                                                                                                                                   | Controller and application version stated for every result                                                                                                              | CDSCO software guidance (21 Jul 2024); software lifecycle and AI change planning                                                                         |
| Status today    | Public datasets only: F1 0.83–0.85 subject-wise, 0.74–0.80 cross-dataset; 0.15–0.18 false alarms per minute; MD-13 test licence for 25 units (10 Aug 2024); ANMS Bhuvanapur proof-of-concept planned | Company reported 41-day comparison; n = 40 vs 42; about 30% vs 14% VAS pain reduction; not peer-reviewed; QMSD and NICE guidelines advise against electrotherapy for OA | IDEAL-01: evaluate devices in stages, with the claim matched to the stage (Bostrom 2014)                                                                 |

|                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>FoGO</b><br>Detection on public data and a bench test study on patients with Parkinson's has yet used it at home. The claim today: 'being developed to detect and cue.' |  <b>Swaknee</b><br>One company reported comparison, not peer-reviewed; QMSD and NICE guidelines advise against the modality. The claim today: 'reported pain reduction in one study.' |  <b>Rule for the Q&amp;A</b><br>Before accepting a stronger objective (validated, proven, safe), ask which row above it reads on, and which device version produced the number. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Sources: P252 BRAC BQ presentation (project-reported); Swagaya evidence page (company-reported); Bala-hin 2010; Neurosurv 2007; Shua de Lima 2017; Tudach 2005; LI, Cochran 2013; QMSD 2019; INICE; NG220; P2 2016; ISO 14971; IEC 62366-1; ISO 14155; CDSCO MDSW guidance 2024; R044-0

Evidence to request before stronger claims

Visual matrix FoGO vs Swaknee  
Interaction —